

The Foundation of Projection Relativity II

Geometric Constraints on the Non-Abelian Phase Fiber

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Abstract

This manuscript extends the projection architecture introduced in Projection Relativity I (PR-I) to the microscopic electroweak and flavor sectors. The central question is whether the particle ledger, usually treated as a collection of independent inputs, can instead be organized as a compact projection structure. We investigate the parent compact non-Abelian phase space

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L,$$

and evaluate the conditions under which this parent fiber locks into the residual electromagnetic boundary. In this formulation, the photon is the unbroken residual projection, while the $W^\pm$ and Z modes arise as massive projection excitations of the broken electroweak directions.

Within this lifted compact fiber, the standard electroweak charge assignments, charged-current ladder structure, neutral-current Z -couplings, Fermi limit, and tree-level relation $\rho_{\text{EW}} = 1$ are recovered as projection-locking consequences. We then explore the boundary deficit as a geometric source for the weak mixing angle and electroweak scale, obtaining a zero-new-parameter closure candidate for $\sin^2 \theta_W$, v_{EW} , and G_F . The same boundary ledger naturally yields a three-generation projection structure,

$$N_{\text{gen}}^{\text{PR}} = \dim[(SU(2)_L \times U(1)_Y)/U(1)_{\text{em}}] = 3.$$

By evaluating sector-specific spectral overlaps across these projection sheets, we obtain charged-lepton, quark, CKM, PMNS, and neutrino mass-sector candidates without using the corresponding target observables as generation inputs. The charged-lepton sector gives a precision Koide-sheet closure candidate, the quark sector yields a compact-boundary hierarchy with PR-derived heavy-quark thresholds and QCD running anchors, and the neutral-lepton sector selects a normal-ordering Majorana-dominant candidate with a boundary prediction for the neutrinoless double-beta effective mass. Throughout, target measurements are used only as diagnostic references after the projection formulas have generated their values. The result is not presented as a completed electroweak precision theorem, but as a test-driven, zero-new-parameter boundary closure program for organizing the electroweak and flavor structure of the Standard Model from projection geometry.

1 Introduction: Compact Electroweak Projection and Flavor Closure

Projection Relativity I (PR-I) [1] introduced a projection-based architecture in which macroscopic gravitational response, displacement-generated inertia, compact electromagnetic phase, finite-core regularization,

and observational diagnostics arise as sectors of a single master projection field,

$$\boxed{\Psi(x, \xi)}. \quad (1)$$

The minimal internal geometry used in PR-I was

$$\boxed{\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1}, \quad (2)$$

where the radial coordinate w supplied the stiffness branch and the compact coordinate θ supplied the residual electromagnetic phase. PR-I fixed the radial projection branch, established the electromagnetic boundary normalization, derived a finite spectral core for strong-field regularization, and provided a public symbolic/numerical verification package for the main macroscopic claims.

The present manuscript begins from that locked PR-I boundary ledger and asks whether the microscopic electroweak and flavor structure can be treated in the same projection language. The working hypothesis is not that the Standard Model [2, 3, 4] is discarded, but that its electroweak and flavor ledger may be reorganized as a compact-boundary projection structure. In particular, PR-II investigates whether weak mixing, the Fermi scale, generation count, flavor hierarchy, CKM mixing [5], PMNS mixing [6], and neutrino mass splittings can be generated from the same spectral data rather than introduced as unrelated numerical inputs. The central extension is the compact electroweak lift

$$\boxed{\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L}. \quad (3)$$

Here S_Y^1 is the compact hypercharge phase and $SU(2)_L$ is the compact non-Abelian weak-isospin fiber. The electromagnetic sector derived in PR-I is not altered. Instead, it is recovered as the residual locked boundary of the parent electroweak fiber:

$$\boxed{SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}, \quad Q = T_3 + \frac{Y}{2}}. \quad (4)$$

In this interpretation, the photon is the unbroken residual projection mode, while $W^\pm$ and Z are massive projection excitation modes associated with the broken electroweak directions.

1.1 From Electroweak Projection to Flavor Ledger

The first part of PR-II establishes the non-Abelian phase projection itself. The $SU(2)_L$ generators obey $[T_a, T_b] = i\epsilon_{abc}T_c$, and the weak charged-current ladder is generated by $T^\pm = T_1 \pm iT_2$.

Projection locking then produces the ordinary electroweak charge assignments, the charged-current topology, the neutral-current Z -couplings, the Fermi limit, and the tree-level relation $\rho_{\text{EW}} = 1$.

The second part of the manuscript asks whether the numerical electroweak scales can be tied to boundary data. The weak mixing angle is treated as a boundary-deficit effect. The electroweak scale is then generated through a spectral hierarchy action built from the PR-I boundary constants and the reduced Planck scale. This produces a zero-new-parameter weak-scale candidate:

$$\boxed{v_{\text{EW}}^{\text{PR}} = 246.219595890 \text{ GeV}}, \quad (5)$$

and

$$\boxed{G_F^{\text{PR}} = 1.166379220 \times 10^{-5} \text{ GeV}^{-2}}. \quad (6)$$

These values are generated before comparison with the observed Fermi scale; the reference measurements are used only afterward as diagnostics.

The third part of PR-II extends the compact projection ledger to flavor. The number of generations is identified with the number of broken compact electroweak projection sheets:

$$N_{\text{gen}}^{\text{PR}} = \dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (7)$$

This provides a geometric origin candidate for the three-generation ledger. Sector-specific overlap operators are then constructed for charged leptons, quarks, and neutrinos. The charged-lepton sector yields a Koide-sheet precision candidate [7]; the quark sector yields heavy-threshold and light-quark hierarchy candidates; and the neutral-lepton sector yields a normal-ordering Majorana-dominant neutrino candidate.

1.2 Scope of the Claims

To keep the status of each result clear, the claims in this manuscript are grouped into four categories.

Structural identities are results that follow directly from the algebra or representation content of the construction. These include the $SU(2)_L$ generator algebra, the residual electromagnetic charge operator

$$Q = T_3 + \frac{Y}{2}, \quad (8)$$

the massless photon eigenmode, the tree-level W/Z mass relations, the Fermi matching condition, the tree-level scalar-coupling ledger, and anomaly cancellation for the chiral fermion representation [8, 9].

Closure candidates are boundary relations proposed to connect the inherited PR-I boundary data to quantities that are usually treated as independent inputs. This category includes the weak-mixing relation, the strong-coupling anchor, the three-generation projection rule, the sector-specific overlap operators, the CKM and PMNS constructions, and the neutrino mass-splitting hierarchy.

Precision candidates are closure relations whose generated values can be compared directly with high-precision reference data. These include the weak-scale construction, the charged-lepton Koide-sheet closure [7], and the heavy-quark threshold-scale candidates. The label “precision candidate” refers to the numerical agreement of the construction; it does not imply that radiative corrections, renormalization-scheme dependence, or higher-order matching have already been completed.

Diagnostic ledgers provide the running, matching, and comparison framework needed to interpret the generated quantities. These include QCD threshold matching, higher-loop running, electroweak running conventions, and comparisons with reference data from the electroweak, flavor, quark, and neutrino sectors. Reference values are introduced only after the PR-II formulas have generated their outputs.

PR-II is therefore not presented as a completed electroweak precision theory. It is presented as a test-driven compact-boundary closure program whose assumptions, generated outputs, diagnostic comparisons, and remaining derivation obligations are stated separately. The central claim is narrower: a substantial part of the tree-level electroweak and flavor ledger can be organized from the PR boundary data without fitting the target observables directly.

1.3 Reproducibility and Test-Driven Method

The development of PR-II follows the same test-driven standard used in PR-I. Each construction is paired with a numerical or symbolic validation harness. The harnesses do not merely confirm the desired equations; they also include negative controls designed to reject common incorrect alternatives, such as Abelianized $SU(2)$, massive photon leakage, wrong Fermi normalization, nonunitary flavor rotations, anomaly-violating fermion ledgers, inverted suppression factors, or hidden use of target observables.

Reproducibility statement. The PR-II electroweak/flavor chain is supported by step-specific validation scripts and more than six hundred automated checks. The harness ledger verifies the compact electroweak lift, projection locking, weak mixing, weak-scale closure, charged and neutral currents, anomaly cancellation, generation replication, flavor hierarchy, charged-lepton precision closure, QCD running, heavy-quark thresholds, CKM and PMNS mixing, and the neutrino mass-sector candidate. Target observables such as G_F , v_{EW} , M_W , M_Z , charged-lepton masses, quark masses, CKM angles, PMNS angles, $\alpha_s(M_Z)$, and neutrino mass splittings are not used as generation inputs. They enter only as diagnostic references. The code and generated outputs are maintained in the public repository: github.com/oshetskieresearch/Projection_Relativity.

2 PR-I Boundary Ledger and the Electroweak Lift

Projection Relativity II begins from the boundary ledger fixed in PR-I. The purpose of this section is not to rederive the macroscopic theory, but to identify the specific PR-I quantities that are inherited by the electroweak and flavor construction. These quantities are treated as fixed geometric data, not as electroweak fit parameters. In PR-I, the minimal internal projection geometry was

$$\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1. \quad (9)$$

The non-compact coordinate w supplied the radial stiffness direction, while the compact coordinate θ supplied the residual electromagnetic phase boundary. The radial stiffness branch was fixed by the projection-trace condition,

$$O_X = -\frac{d^2}{dw^2} + 1 + w^2 + \frac{3}{4}w^4. \quad (10)$$

The corresponding radial gap is

$$\mu_{\text{min}}^2 = 3.052966743096. \quad (11)$$

This radial gap is part of the strong-field and finite-core sector. It is not refit in the electroweak construction. The compact electromagnetic boundary from PR-I is encoded in the boundary-resolved compact stiffness

$$q_{\text{bc}} = 10.905007182855176, \quad (12)$$

and the boundary cofactor

$$c_{\text{bc}} = \frac{3354902985}{4211081216} = 0.796684464847899. \quad (13)$$

Together they reproduce the residual compact electromagnetic normalization¹,

$$\boxed{\alpha_{\text{PR,bc}}^{-1} = 4\pi q_{\text{bc}} = 137.036361812007.} \quad (14)$$

This compact electromagnetic boundary is preserved in PR-II. The weak sector is not introduced by altering the PR-I electromagnetic result. Instead, PR-II asks whether that residual $U(1)_{\text{em}}$ boundary is the locked remnant of a larger compact electroweak fiber. The determinant suppression ratio inherited from PR-I is

$$r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}} = 0.073056757459128. \quad (15)$$

¹**Note on numerical resolution and framework scope.** The numerical constants used throughout PR-II are inherited from the published finite-resolution PR-I ledger so that every downstream result remains reproducible from the original release. Independent banded harmonic-basis calculations stabilize the low radial spectrum and finite-rank boundary map at $\mu_{\text{min}}^2 \simeq 3.053016666733$, $\alpha_{\text{PR,bc}}^{-1} \simeq 137.036041314016$. This reduces the electromagnetic residual from approximately 2.65 ppm in the original ledger to approximately 0.31 ppm relative to the CODATA recommended value. The refinement demonstrates numerical stability and substantially improved agreement. The refined constants are not mixed into the present PR-II calculations. Their consistent propagation is reserved for a separate high-resolution release.

This ratio will appear repeatedly in the flavor ledger as the compact generation-sheet suppression factor. The residual compact boundary deficit is

$$\delta_{bc} = 1 - c_{bc} = 0.203315535152101. \quad (16)$$

The normalized boundary deficit is

$$\varepsilon_{bc} = \frac{1 - c_{bc}}{q_{bc}} = 0.018644236701811. \quad (17)$$

At this stage, ε_{bc} is only an inherited compact boundary quantity. Subsequent sections test whether this normalized boundary deficit fixes the weak mixing angle through the compact electroweak projection.

Table 1: PR-I boundary quantities inherited by PR-II.

Quantity	Value	Role in PR-II
$\mu_{\min}^2$	3.052966743096	Radial stiffness gap inherited from PR-I.
q_{bc}	10.905007182855176	Compact boundary stiffness.
c_{bc}	0.796684464847899	boundary cofactor.
$\alpha_{\text{PR},bc}^{-1}$	137.036361812007	Residual electromagnetic boundary normalization.
$r_G = c_{bc}/q_{bc}$	0.073056757459128	Compact generation-sheet suppression ratio.
$\varepsilon_{bc} = (1 - c_{bc})/q_{bc}$	0.018644236701811	Normalized compact boundary deficit.

2.1 Electroweak Lift of the Compact Phase Sector

The PR-II microscopic extension lifts the compact phase sector from the residual electromagnetic boundary into a parent compact electroweak fiber,

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L. \quad (18)$$

Here S_Y^1 is the compact hypercharge phase and $SU(2)_L$ is the compact non-Abelian weak-isospin fiber. The radial factor $\mathbb{R}_w$ and its stiffness branch are inherited from PR-I. The new structure is the non-Abelian compact phase factor $SU(2)_L$ [10, 4]. The corresponding internal electroweak operator is written schematically as

$$O_{\text{int}}^{\text{EW}} = O_X + O_Y + O_L. \quad (19)$$

The radial operator is the PR-I stiffness operator,

$$O_X = -\frac{d^2}{dw^2} + 1 + w^2 + \frac{3}{4}w^4. \quad (20)$$

The compact hypercharge operator is

$$O_Y = -\frac{1}{R_Y^2} \frac{d^2}{d\theta_Y^2}, \quad (21)$$

and the compact weak-isospin operator is

$$O_L = -\frac{1}{R_L^2} \Delta_{SU(2)}. \quad (22)$$

The $SU(2)_L$ compact fiber carries non-Abelian generators T_a satisfying

$$[T_a, T_b] = i\epsilon_{abc} T_c. \quad (23)$$

In the fundamental representation,

$$T_a = \frac{\sigma_a}{2}, \quad (24)$$

where σ_a are the Pauli matrices. The charged-current compact ladder operators are $T^\pm = T_1 \pm iT_2$. This non-Abelian ladder structure is the first microscopic distinction between the weak compact fiber and the residual electromagnetic boundary. The compact weak-isospin spectrum is generated by the $SU(2)$ Laplacian,

$$O_L D_{mn}^j = \frac{j(j+1)}{R_L^2} D_{mn}^j. \quad (25)$$

Thus the compact weak spectrum is

$$\lambda_j^{(L)} = \frac{j(j+1)}{R_L^2}. \quad (26)$$

The singlet $j = 0$ is weak-isospin neutral, the doublet $j = 1/2$ supplies the weak matter representation, and the adjoint $j = 1$ supplies the weak projection modes. The operational diagram of the relationship from PR-I internal geometry to PR-II electroweak lift is illustrated in Figure 1

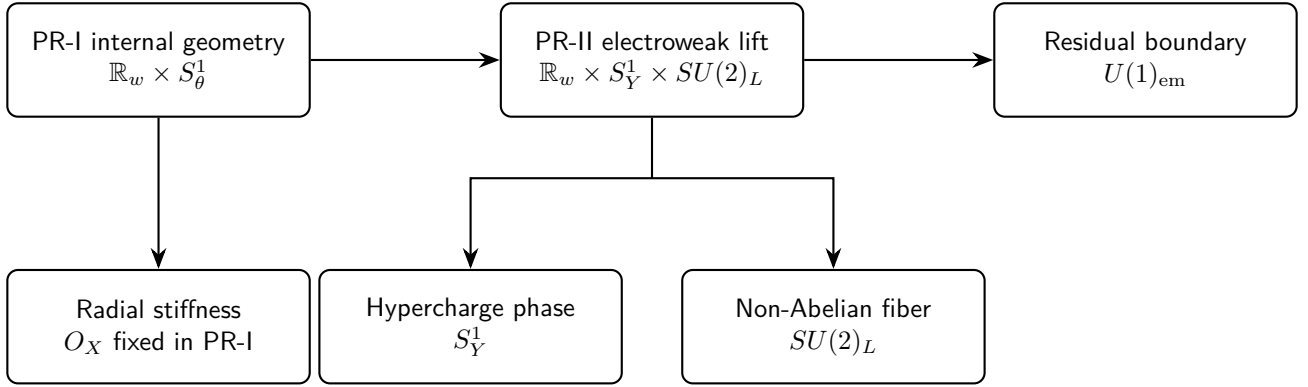


Figure 1: PR-II compact electroweak lift preserves the PR-I electromagnetic boundary. The radial stiffness is inherited from the macroscopic theory, while the non-Abelian fiber and hypercharge phase are introduced to govern the microscopic sector.

2.2 Residual Electromagnetic Boundary

The residual electromagnetic boundary is selected via projection locking, reducing the parent electroweak symmetry [11, 12]:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \quad (27)$$

The resulting unbroken generator dictates the residual charge:

$$Q = T_3 + \frac{Y}{2}. \quad (28)$$

We note that this is the exact same electromagnetic boundary whose compact normalization was established in PR-I. Rather than discarding or replacing the PR-I electromagnetic sector, this framework reinterprets it: electromagnetism is simply the massless residual projection that survives after the parent compact electroweak fiber locks.

Applying this generator naturally recovers standard charge assignments. For a left-handed lepton doublet with $Y = -1$, the matrix evaluates to:

$$Q = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \quad (29)$$

Similarly, for a left-handed quark doublet with $Y = 1/3$:

$$Q = \begin{pmatrix} 2/3 & 0 \\ 0 & -1/3 \end{pmatrix}. \quad (30)$$

Consequently, the conventional electromagnetic charges emerge from the compact projection-locking relation as described in equation 28. Translated into the language of PR-II, the photon operates as the unbroken residual compact projection mode. Furthermore, the $W^\pm$ and Z modes are no longer treated as unrelated, primitive objects; instead, they manifest as massive projection excitation modes associated with the broken directions of the parent electroweak fiber. This interpretation preserves the standard low-energy electroweak ledger, replacing independent parameter insertion, where possible, with compact-boundary closure tests.

2.3 Input Discipline

The purpose of PR-II is to test whether the electroweak and flavor ledger can be generated from compact projection data without introducing new fitted microscopic parameters. Accordingly, the following quantities are not allowed as generation inputs in the closure chain [13, 14]:

Table 2: The forbidden generation inputs. These quantities may be used only after a PR-II formula has generated its value, as diagnostic reference data.

Physics Sector	Excluded Target Observables
Electroweak Scale & Bosons	G_F, v_{EW}, M_W, M_Z
Charged Lepton Masses	m_e, m_μ, m_τ
Quark Masses	$m_u, m_d, m_s, m_c, m_b, m_t$
Flavor Mixing & QCD Anchors	$V_{CKM}, U_{PMNS}, \alpha_s(M_Z), \Lambda_{QCD}$
Neutrino Mass Sector	$\Delta m_{21}^2, \Delta m_{31}^2, \sum_i m_i$

The starting data allowed are the PR-I boundary ledger, the compact group structure, and the fundamental constants. In particular, PR-II may only use starting points as summarized in Table 3. This input discipline is central to the manuscript. Whenever a result is compared to an electroweak, flavor, quark, or neutrino measurement, that measurement must be entered only as a diagnostic reference. If a proposed closure requires importing a target observable as an input, it is not a zero-new-parameter PR-II closure. It must instead be classified as a diagnostic ledger or incomplete sector.

3 Compact Non-Abelian Phase Projection

The electroweak lift introduced the parent compact fiber

$$\boxed{\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L} \quad (31)$$

Table 3: The PR-II Allowed Input Ledger. These are the permitted geometric and topological starting inputs; target observables are forbidden from this list.

PR Term	Description	Origin	Value
q_{bc}	Compact boundary stiffness	PR-I EM Boundary	10.905007182855176
c_{bc}	Compact boundary cofactor	PR-I EM Boundary	0.796684464847899
r_G	Compact generation suppression	PR-I Ratio (c_{bc}/q_{bc})	0.073056757459128
ε_{bc}	Normalized boundary deficit	PR-I Deficit ($((1 - c_{bc})/q_{bc})$)	0.018644236701811
$\mu_{\min}^2$	Radial stiffness gap	PR-I Radial Operator	3.052966743096
T_3	Weak-isospin generator	$SU(2)_L$ Compact Fiber	Rep. dependent
Y	Hypercharge generator	$U(1)_Y$ Compact Fiber	Rep. dependent
$C_2^{SU(2)}$	Weak-isospin Casimir	$SU(2)_L$ Group Topology	3/4 (doublets)
$C_2^{SU(3)}$	Color Casimir	$SU(3)_c$ Group Topology	4/3 (triplets)
$\bar{M}_{Pl}$	Reduced Planck mass	Fundamental Constants	$\approx 2.435 \times 10^{18}$ GeV

This section isolates the fundamental microscopic extension introduced by PR-II; the compact non-Abelian phase projection carried by $SU(2)_L$. The residual electromagnetic boundary inherited from PR-I is Abelian, with its compact phase governed by a single commuting generator. The weak sector, by contrast, is not merely a secondary copy of the electromagnetic phase. It is a non-Abelian compact phase fiber, characterized by internal orientation, generator non-commutativity, a distinct charged-current ladder structure, and a discrete compact spectral hierarchy.

3.1 Group-Valued Internal Phase

In PR-I, the compact electromagnetic phase was represented by a circle coordinate, $\theta \in S^1$. The corresponding compact phase factor was Abelian $e^{i\theta} \in U(1)$. In PR-II, the weak phase is instead group-valued:

$$\boxed{g_L \in SU(2)_L.} \quad (32)$$

The master projection field may therefore be written schematically as

$$\boxed{\Psi = \Psi(x, w, \theta_Y, g_L),} \quad (33)$$

where x denotes observable spacetime coordinates, w is the radial stiffness coordinate, θ_Y is the compact hypercharge phase, and g_L is the compact weak-isospin orientation. The weak projection Hilbert space is

$$\mathcal{H}_L = L^2(SU(2)_L, d\mu_H), \quad (34)$$

where $d\mu_H$ is the Haar measure on $SU(2)$. By the Peter–Weyl decomposition, square-integrable functions on $SU(2)$ decompose into matrix elements of irreducible representations:

$$\mathcal{H}_L = \bigoplus_{j=0, \frac{1}{2}, 1, \dots} \mathcal{H}_j \otimes \mathcal{H}_j^*. \quad (35)$$

A convenient basis is given by the Wigner matrix elements,

$$D_{mn}^j(g_L), \quad m, n = -j, \dots, j. \quad (36)$$

So the weak compact fiber naturally carries singlet, doublet, and adjoint spectral sectors:

$$j = 0, \quad j = \frac{1}{2}, \quad j = 1. \quad (37)$$

3.2 Non-Abelian Generator Algebra

The $SU(2)_L$ generators are denoted T_a , with $a = 1, 2, 3$. They obey the non-Abelian algebra $[T_a, T_b] = i\epsilon_{abc}T_c$. In the fundamental representation,

$$T_a = \frac{\sigma_a}{2}, \quad (38)$$

where σ_a are the Pauli matrices. The normalization is

$$\text{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}. \quad (39)$$

The fundamental Casimir is

$$\boxed{\sum_{a=1}^3 T_a^2 = \frac{3}{4}I.} \quad (40)$$

This identifies the weak doublet as the $j = 1/2$ compact spectral representation. The adjoint weak projection modes transform in the $j = 1$ representation. The adjoint Casimir is

$$C_2(j = 1) = 1(1 + 1) = 2. \quad (41)$$

Therefore the compact weak fiber distinguishes the weak matter doublet from the weak projection modes by representation class:

$$\boxed{j = \frac{1}{2} \quad \text{for weak matter doublets,} \quad j = 1 \quad \text{for weak projection modes.}} \quad (42)$$

3.3 Compact Weak-Isospin Operator

The compact weak-isospin operator is the $SU(2)$ Laplacian,

$$O_L = -\frac{1}{R_L^2}\Delta_{SU(2)}. \quad (43)$$

Acting on the Wigner basis, it gives

$$O_L D_{mn}^j = \frac{j(j+1)}{R_L^2} D_{mn}^j. \quad (44)$$

The compact weak spectral values are therefore

$$\boxed{\lambda_j^{(L)} = \frac{j(j+1)}{R_L^2}.} \quad (45)$$

The first three sectors are

$$\boxed{\lambda_0^{(L)} = 0, \quad \lambda_{1/2}^{(L)} = \frac{3}{4R_L^2}, \quad \lambda_1^{(L)} = \frac{2}{R_L^2}.} \quad (46)$$

In PR-II language, the $j = 0$ sector is weak-isospin neutral, the $j = 1/2$ sector supplies the compact weak doublet ledger, and the $j = 1$ sector supplies the compact weak projection modes. This spectral hierarchy is the microscopic origin of the weak-sector distinction between matter doublets and gauge excitation directions. Mass does not appear at this stage. Massive $W^\pm$ and Z modes arise only after projection locking, when the parent compact electroweak fiber selects a residual electromagnetic boundary.

3.4 Charged-Current Topological Ladder

The charged-current topological ladder is generated by

$$T^\pm = T_1 \pm iT_2. \quad (47)$$

In the fundamental doublet basis,

$$|\text{up}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\text{down}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (48)$$

The ladder operators act as $T^+|\text{down}\rangle = |\text{up}\rangle, T^-|\text{up}\rangle = |\text{down}\rangle$. The complementary actions vanish $T^+|\text{up}\rangle = 0, T^-|\text{down}\rangle = 0$. The weak charged-current transitions are compact-isospin ladder moves within a weak doublet. This is the PR-II interpretation of weak charged-current topology. A weak transition is not inserted as a separate rule. It is the motion of a state across the compact non-Abelian phase fiber:

$$\boxed{|\text{down}\rangle \xrightarrow{T^+} |\text{up}\rangle, \quad |\text{up}\rangle \xrightarrow{T^-} |\text{down}\rangle.} \quad (49)$$

After projection locking, these ladder moves become the charged-current weak processes mediated by the massive projection excitation modes $W^\pm$.

3.5 Hypercharge Commutation and Electroweak Product Structure

The electroweak parent fiber is a product,

$$SU(2)_L \times U(1)_Y. \quad (50)$$

The hypercharge generator Y commutes with the weak-isospin generators:

$$[Y, T_a] = 0. \quad (51)$$

Therefore, the compact hypercharge phase and weak non-Abelian phase are independent before projection locking. The parent electroweak connection may be written as

$$\mathcal{A}_\mu^{\text{EW}} = gW_\mu^a T_a + g' B_\mu \frac{Y}{2}. \quad (52)$$

The covariant derivative is

$$D_\mu = \partial_\mu - igW_\mu^a T_a - ig' B_\mu \frac{Y}{2}. \quad (53)$$

The non-Abelian field strength contains the commutator term

$$F_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g\epsilon^{abc} W_\mu^b W_\nu^c. \quad (54)$$

This term is absent in an Abelian compact phase. It is the curvature expression of the compact non-Abelian phase projection. The hypercharge curvature remains Abelian:

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (55)$$

The parent compact electroweak fiber therefore contains both an Abelian compact phase and a non-Abelian compact phase. The residual electromagnetic boundary appears only after these sectors lock into the unbroken generator as described in equation 28.

3.6 Structural Status of the Weak Projection Modes

Within this lifted compact fiber, rather than introducing independent fundamental gauge fields, the internal geometry of $SU(2)_L$ supplies three distinct weak projection directions, $W_\mu^1, W_\mu^2, W_\mu^3$. The charged combinations are

$$\boxed{W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2).} \quad (56)$$

The neutral direction W_μ^3 does not by itself equal the physical Z or the photon. The physical neutral projection modes are obtained only after W_μ^3 mixes with the hypercharge projection mode B_μ . That mixing belongs to the projection-locking step. So that the status of the weak projection modes at this point is structural:

$$\boxed{SU(2)_L \text{ provides the compact non-Abelian phase directions.}} \quad (57)$$

The later projection-locking condition determines which combination remains massless and which combinations become massive projection excitation modes.

3.7 Negative Structural Boundary

The compact weak sector cannot be consistently replaced by three independent Abelian phases. If the commutator is removed,

$$[T_a, T_b] = 0, \quad (58)$$

then the ladder structure, adjoint curvature term, and compact non-Abelian projection geometry disappear. Such an Abelianized structure cannot support the charged-current topological ladder or the standard $SU(2)_L$ representation ledger. Thus the first falsification boundary of the PR-II weak sector is structural:

$$\boxed{\text{weak projection} \neq U(1) \times U(1) \times U(1).} \quad (59)$$

The weak sector must retain

$$\boxed{[T_a, T_b] = i\epsilon_{abc}T_c.} \quad (60)$$

This non-Abelian compact algebra provides the minimal microscopic structure required to support charged-current topology, weak doublets, and the subsequent W/Z mass ledger. With the parent compact electroweak fiber now formally established, the following section introduces the mechanics of projection locking. This process demonstrates how the parent fiber preserves the residual electromagnetic boundary, naturally yielding one massless photon alongside three massive weak projection excitation modes.

4 Projection Locking and the Electroweak Gauge Ledger

The compact non-Abelian phase projection serves as the foundation for the parent electroweak fiber, $SU(2)_L \times U(1)_Y$. The immediate focus is to determine the residual electromagnetic boundary and the massive weak projection modes generated upon the locking of this parent fiber. This projection-locking step is constrained by three strict requirements. First, one neutral mode must remain exactly massless. Second, the unbroken residual generator must reproduce the standard electromagnetic charge as defined in equation 28. Third, the remaining broken compact directions must generate the massive $W^\pm$ and Z projection modes.

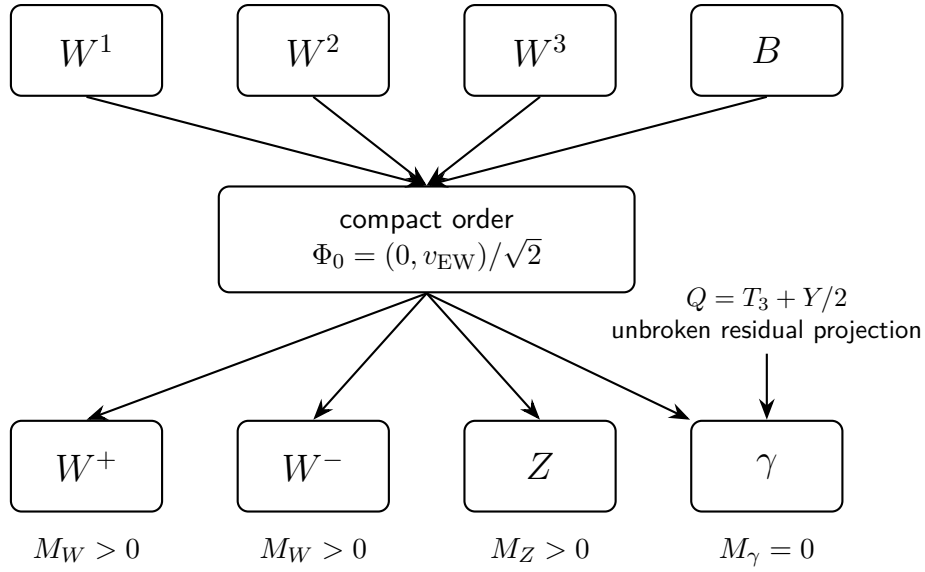


Figure 2: Projection locking maps the parent electroweak modes (W^1, W^2, W^3, B) into the physical massive weak modes ($W^\pm, Z$) and the massless residual photon (γ). The photon remains massless because it constitutes the unbroken $Q = T_3 + Y/2$ projection mode.

4.1 Compact Electroweak Order Direction

Projection locking is represented by a compact electroweak order direction. For the minimal doublet locking direction [11],

$$\Phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{EW} \end{pmatrix}, \quad Y = 1. \quad (61)$$

The weak-isospin generator T_3 acts on the lower component as

$$T_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (62)$$

So that we find,

$$Q\Phi_0 = \left(T_3 + \frac{Y}{2}\right)\Phi_0 = 0. \quad (63)$$

The locked direction is neutral under Q . Thus Q generates the residual electromagnetic boundary:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \quad (64)$$

This equation is the projection-locking statement. The compact electromagnetic boundary fixed in PR-I is not discarded. It is recovered as the unbroken residual projection of the parent electroweak compact fiber.

4.2 Electroweak Connection and Mass Ledger

The physical kinematics of the compact non-Abelian fiber are governed by the parent electroweak connection. As the parent fiber locks into its residual electromagnetic boundary, the geometric constraints on the covariant derivative generate a mass matrix for the excitation modes. This section demonstrates how that locking mechanism defines the massive weak modes while guaranteeing the photon remains massless.

The parent compact electroweak connection using equation 52 and 53. The projection-locking mass ledger comes from the compact order term

$$(D_\mu \Phi_0)^\dagger (D^\mu \Phi_0). \quad (65)$$

The charged projection modes are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (66)$$

Their tree-level mass is

$$M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4}. \quad (67)$$

The neutral modes W_μ^3 and B_μ mix through the mass matrix

$$M_0^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}. \quad (68)$$

The determinant vanishes $\det M_0^2 = 0$, so that one neutral projection mode remains massless $M_\gamma = 0$. This massless mode is the residual electromagnetic projection mode inherited from PR-I. The massive neutral eigenvalue is

$$M_Z^2 = \frac{(g^2 + g'^2)v_{\text{EW}}^2}{4}. \quad (69)$$

So that,

$$\boxed{\text{spec}(M_{\text{EW}}^2) = \{0, M_W^2, M_W^2, M_Z^2\}}. \quad (70)$$

Projection locking therefore produces one massless residual electromagnetic mode and three massive weak projection excitation modes.

4.3 Neutral Mixing and the Residual Photon

The vanishing determinant of the neutral mass matrix dictates that the physical states must be orthogonal combinations of the W_μ^3 and B_μ projection directions. To isolate the unbroken electromagnetic phase from the massive weak sector, these modes are rotated into the physical mass basis. This geometric diagonalization separates the massless residual photon from the massive Z excitation, parameterized by the weak mixing angle. The neutral mass eigenstates are:

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu. \quad (71)$$

The weak mixing angle satisfies

$$\tan \theta_W = \frac{g'}{g}. \quad (72)$$

So that we find,

$$\boxed{\sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}}, \quad \cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}}. \quad (73)$$

The residual electromagnetic coupling is $e = g \sin \theta_W = g' \cos \theta_W$. The photon A_μ is the unbroken Q -projection mode. The Z_μ is the neutral projection excitation orthogonal to the residual electromagnetic boundary. The neutral projection-locking split is $(W_\mu^3, B_\mu) \rightarrow (A_\mu, Z_\mu)$, with $M_A = 0, M_Z > 0$.

4.4 Tree-Level Electroweak Consistency

To validate the projection-locking mechanism, it must be shown to faithfully reproduce the established phenomenological bounds of the low-energy effective Standard Model. The immediate test of this alignment is the relationship between the charged and neutral excitation masses [14], which are geometrically constrained to obey:

$$M_W = M_Z \cos \theta_W. \quad (74)$$

Consequently, the tree-level electroweak rho parameter reduces to unity:

$$\rho_{EW} = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1 \quad (75)$$

This identity is a structural consequence of the compact doublet locking direction; it is not an engineered numerical fit. This same structural rigidity guarantees the correct Fermi contact limit once the massive charged projection mode is integrated out at low momentum transfer, $q^2 \ll M_W^2$. At this geometric stage, the matching relation is:

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} = \frac{1}{2v_{EW}^2}. \quad (76)$$

4.5 Compact Boundary Closure for the Weak Mixing Angle

The weak mixing angle is the first electroweak quantity tested against the inherited PR-I compact boundary data. The raw compact hypercharge trace fraction is

$$T_Y = \frac{1}{4}. \quad (77)$$

The compact boundary cofactor and boundary stiffness inherited from PR-I are $c_{bc} = 0.796684464847899$, $q_{bc} = 10.905007182855176$. The normalized compact boundary deficit is

$$\Delta_{EW} = \frac{1 - c_{bc}}{q_{bc}}. \quad (78)$$

Numerically,

$$\Delta_{EW} = 0.018644236701811. \quad (79)$$

The PR-II compact-boundary weak-mixing candidate is

$$\sin^2 \theta_W^{PR} = T_Y - \Delta_{EW}. \quad (80)$$

Equivalently,

$$\sin^2 \theta_W^{PR} = \frac{1}{4} - \frac{1 - c_{bc}}{q_{bc}}. \quad (81)$$

Substitution gives

$$\sin^2 \theta_W^{PR} = 0.231355763298189. \quad (82)$$

We also find,

$$\cos^2 \theta_W^{PR} = 0.768644236701811, \quad (83)$$

and

$$\frac{g'}{g} = \tan \theta_W^{PR} = 0.548627374785032. \quad (84)$$

This result should be interpreted as a compact-boundary weak-mixing closure candidate. It does not yet include radiative electroweak running, scheme dependence, or full precision-observable matching. The structural claim is that the weak mixing angle is not inserted as a new free parameter. It is generated by subtracting the normalized compact boundary deficit from the raw hypercharge trace fraction.

4.6 Residual Charge Assignments

It is necessary to verify that its unbroken geometric generator correctly governs the matter sector. The electromagnetic charges are derived from the projection-locking relation. The unbroken residual generator is dictated by equation 28. Evaluating this unbroken direction naturally recovers the standard Standard Model charge assignments [14]. For the left-handed lepton doublet, the residual electromagnetic generator recovers the standard charge assignments. For the left-handed lepton doublet,

$$L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Y = -1, \quad (85)$$

we obtain

$$Q_\nu = 0, \quad Q_e = -1. \quad (86)$$

For the left-handed quark doublet,

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad Y = \frac{1}{3}, \quad (87)$$

we obtain

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}. \quad (88)$$

Right-handed singlets have $T_3 = 0$, so their electromagnetic charge is fixed by

$$Q = \frac{Y}{2}. \quad (89)$$

Therefore the residual electromagnetic boundary simultaneously preserves the PR-I compact electromagnetic sector and reproduces the electroweak charge ledger:

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}, \quad Q = T_3 + \frac{Y}{2}. \quad (90)$$

4.7 Projection-Locking Interpretation

Projection locking converts the parent compact electroweak fiber into a residual electromagnetic boundary plus massive weak projection excitations:

$$SU(2)_L \times U(1)_Y \longrightarrow \{U(1)_{\text{em}}, W^+, W^-, Z\}. \quad (91)$$

In PR-II language, $W^\pm$ and Z are not primitive independent additions to the theory. They are the massive excitation modes of the compact electroweak fiber after the residual electromagnetic boundary has been selected. The photon remains massless because it is the unbroken Q -projection mode. The projection-locking chain is therefore

$$\mathcal{M}_{\text{int}}^{\text{EW}} \rightarrow SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}} \rightarrow \{M_\gamma = 0, M_W, M_Z\}. \quad (92)$$

Including the weak-mixing closure,

$$\boxed{\{q_{bc}, c_{bc}, T_Y\} \rightarrow \sin^2 \theta_W^{\text{PR}} \rightarrow \{A_\mu, Z_\mu, W^\pm\}}. \quad (93)$$

4.8 Falsification Bounds

The validity of the projection-locking mechanism rests on its geometric rigidity; it does not admit arbitrary deformations. This ledger possesses explicit failure modes that falsify the architecture if violated. First, if the neutral mass matrix lacked the strict non-Abelian W^3/B mixing term, taking the decoupled form

$$M_{0,\text{wrong}}^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & 0 \\ 0 & g'^2 \end{pmatrix}, \quad (94)$$

the determinant would be non-zero:

$$\det M_{0,\text{wrong}}^2 \neq 0, \quad (95)$$

resulting in a massive photon. Such a disconnected structure is rejected by the compact electroweak lift. Second, if the unbroken residual generator deviated from the precise locking direction equation 28, the locked vacuum would fail to be neutral, and the standard electromagnetic charge assignments would not be recovered, instantly breaking the low-energy effective ledger. Third, if the weak mixing angle were dictated by the raw hypercharge trace,

$$\sin^2 \theta_W = \frac{1}{4}, \quad (96)$$

it would imply a unbroken compact boundary, entirely ignoring the boundary deficit established in PR-I. Instead, PR-II closure mandates the geometric correction:

$$\sin^2 \theta_W^{\text{PR}} = \frac{1}{4} - \frac{1 - c_{bc}}{q_{bc}}. \quad (97)$$

The projection-locking step is falsifiable at the structural level. To remain viable, the parent fiber must produce a massless photon, standard residual charge assignments, massive $W^\pm$ and Z excitation modes, the tree-level identity $\rho_{\text{EW}} = 1$, and a weak mixing angle dictated by the compact boundary deficit. The failure of any single condition breaks the PR-II electroweak lift. With the projection-locking and electroweak gauge ledger established at the structural level, the subsequent section derives the weak scale and Fermi constant directly from compact-boundary hierarchy closure, rather than treating v_{EW} and G_F as independent input parameters.

5 Weak Mixing, Weak Scale, and Fermi Closure

The completion of the projection-locking step permanently fixed the residual electromagnetic generator,

$$\boxed{Q = T_3 + \frac{Y}{2}}, \quad (98)$$

while simultaneously yielding the compact-boundary weak-mixing candidate:

$$\boxed{\sin^2 \theta_W^{\text{PR}} = 0.231355763298189}. \quad (99)$$

The theoretical burden now shifts to the mass scale of the weak sector. Specifically, we must determine whether the electroweak scale can be derived from the existing compact-boundary ledger, thereby eliminating the need to import the Fermi constant as a free parameter. This section executes that weak-scale

closure in three sequential steps. First, the phenomenological Fermi ledger is established as a necessary low-energy consistency relation. Next, the dimensionless compact weak-order factor is constructed from the inherited PR-I boundary constants. Third, the weak scale is extracted directly from a compact spectral hierarchy action.

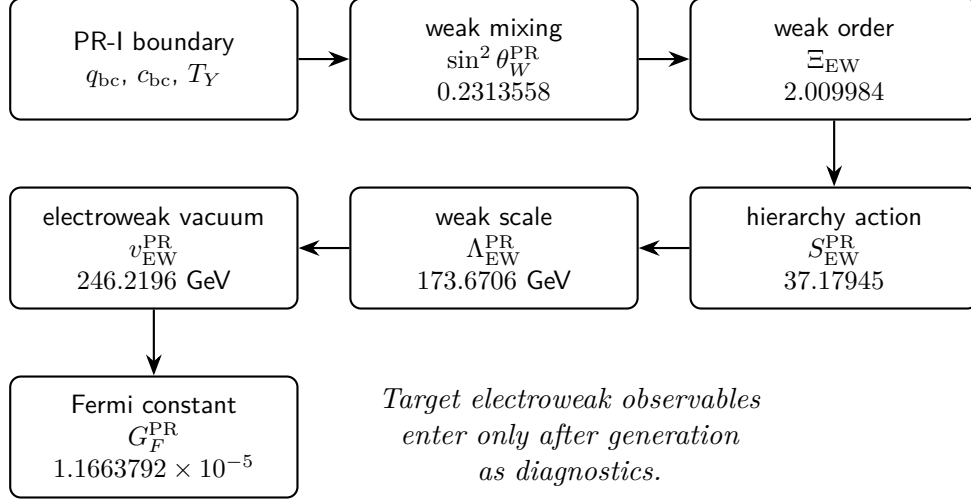


Figure 3: The compact-boundary hierarchy chain for the weak scale. The derivation moves sequentially from the inherited geometric boundary data down to the fundamental electroweak parameters, completely isolating them from empirical input fitting.

5.1 Fermi Consistency Ledger

Before generating the weak scale from the compact-boundary geometry, it is first necessary to define the precise phenomenological target it must satisfy. In the standard low-energy limit, the weak scale is parameterized by the Fermi constant [15], which emerges as a residual contact interaction. Specifically, at low momentum transfer, $q^2 \ll M_W^2$, integrating out the massive charged weak projection mode naturally produces this contact interaction. The invariant matching condition is

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (100)$$

Applying the charged weak mass relation derived from the compact order,

$$M_W^2 = \frac{g^2 v_{EW}^2}{4}, \quad (101)$$

the Fermi relation firmly reduces to

$$\frac{G_F}{\sqrt{2}} = \frac{1}{2v_{EW}^2}. \quad (102)$$

Equivalently, the standard electroweak vacuum scale is defined by

$$v_{EW} = \left(\sqrt{2} G_F \right)^{-1/2}. \quad (103)$$

Crucially, within the PR-II framework, this relation is a low-energy consistency condition rather than a mechanism to insert the weak scale by hand. It explicitly defines the exact phenomenological target that the subsequent compact-boundary derivation of v_{EW}^{PR} must recover.

5.2 Dimensionless Weak-Order Factor

The geometric generation of the weak scale requires separating its dimensionless geometric scaling factor from its fundamental energy scale. This dimensionless order factor is completely determined by the inherited PR-I boundary constants and the weak mixing angle. The boundary weak-mixing closure provides:

$$\boxed{\sin^2 \theta_W^{\text{PR}} = T_Y - \Delta_{\text{EW}},} \quad (104)$$

where

$$\boxed{T_Y = \frac{1}{4}, \quad \Delta_{\text{EW}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}}.} \quad (105)$$

Using the PR-I boundary values, $q_{\text{bc}}, c_{\text{bc}}$ gives

$$\boxed{\Delta_{\text{EW}} = 0.018644236701811.} \quad (106)$$

The dimensionless electroweak order factor is defined by

$$\boxed{\Xi_{\text{EW}} = q_{\text{bc}} c_{\text{bc}} \sin^2 \theta_W^{\text{PR}}.} \quad (107)$$

This centralized quantity encapsulates the compact boundary stiffness, the compact boundary cofactor, and the weak-mixing projection fraction. Direct substitution yields the numerical values:

$$\boxed{\Xi_{\text{EW}} = 2.0099841245909515, \quad \sqrt{\Xi_{\text{EW}}} = 1.417739089039641.} \quad (108)$$

By factoring out this dimensionless geometric weight, the weak order scale is written as

$$\boxed{v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}},} \quad (109)$$

where $\Lambda_{\text{EW}}^{\text{PR}}$ serves as the terminal dimensionful weak displacement anchor. The remaining imperative task is to generate this scale $\Lambda_{\text{EW}}^{\text{PR}}$ directly from the compact geometry.

5.3 Terminal Weak Displacement Overlap

To complete the electroweak derivation, we must bridge the gap between the dimensionless order factor and the displacement scale. This is achieved through the terminal weak matter–displacement overlap, which anchors the vacuum scale to the compact hierarchy. This relation reveals that the overlap is near unity, effectively coupling the electroweak vacuum to the geometric displacement of the parent fiber. The terminal weak matter–displacement overlap is defined by:

$$\left(y_T^{\text{PR}}\right)^2 \Xi_{\text{EW}} = 2. \quad (110)$$

So that we find,

$$\boxed{y_T^{\text{PR}} = \sqrt{\frac{2}{\Xi_{\text{EW}}}} = 0.997513275401799.} \quad (111)$$

The terminal weak displacement anchor is related to the weak order scale by

$$\Lambda_{\text{EW}}^{\text{PR}} = \frac{y_T^{\text{PR}} v_{\text{EW}}^{\text{PR}}}{\sqrt{2}}. \quad (112)$$

Equivalently,

$$\Lambda_{\text{EW}}^{\text{PR}} = \frac{v_{\text{EW}}^{\text{PR}}}{\sqrt{\Xi_{\text{EW}}}} \quad (113)$$

This relation shows that the terminal weak displacement overlap is near unity. In PR-II language, the weak scale is therefore tied to a terminal matter-displacement channel whose dimensionless overlap is fixed by compact-boundary data:

$$\boxed{y_T^{\text{PR}} \simeq 1.} \quad (114)$$

This relation reveals that the terminal weak displacement overlap is geometrically constrained to be nearly unity, reflecting the high-degree alignment of the electroweak vacuum with the matter-displacement channel.

5.4 Compact Spectral Hierarchy Action

The large disparity between the reduced Planck mass and the electroweak scale finds a natural, deterministic resolution within the PR-II framework [16]. Instead of requiring parameter tuning, the weak scale emerges as an exponentially suppressed projection of the gravitational scale, governed by the geometry of the compact boundary. We define this suppression through a hierarchy action constructed entirely from the foundational boundary constants established in the preceding sections. This derivation yields a zero-new-parameter prediction for the weak displacement anchor, demonstrating that the electroweak scale is an inherent consequence of the compact spectral topology. The natural dimensional reference scale for a zero-new-parameter weak-scale closure is the reduced Planck mass [13],

$$\boxed{\bar{M}_{\text{Pl}} = \sqrt{\frac{\hbar c}{8\pi G_N}} = 2.4353234593382036 \times 10^{18} \text{ GeV}.} \quad (115)$$

The electroweak scale must then arise from a dimensionless spectral suppression:

$$\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} \exp\left(-S_{\text{EW}}^{\text{PR}}\right). \quad (116)$$

The leading compact-boundary hierarchy action is

$$S_{\text{EW}}^{\text{PR,base}} = \pi q_{\text{bc}} + \ln\left(\frac{q_{\text{bc}}}{c_{\text{bc}}}\right) + \ln\left(\frac{4}{3}\right) + \left(T_Y - \sin^2 \theta_W^{\text{PR}}\right) \frac{1 + c_{\text{bc}}}{2}. \quad (117)$$

Each term has a fixed geometric role. The first term is the radial compact-boundary action. The second is the compact stiffness/cofactor determinant ratio. The third is the 3+1 projection-trace orientation factor. The final term is the weak boundary-deficit correction averaged over the residual compact boundary. The leading action gives

$$\boxed{S_{\text{EW}}^{\text{PR,base}} = 37.18004007116382.} \quad (118)$$

At leading order this already produces the weak displacement anchor to the sub-per-mille level. We introduce the following second-order compact-boundary closure term

$$\delta S_{\text{EW}}^{(2)} = -\frac{1}{2} (1 + 3c_{\text{bc}}) \Delta_{\text{EW}}^2. \quad (119)$$

The factor

$$A_{\text{EW}} = \frac{1 + 3c_{\text{bc}}}{2} \quad (120)$$

combines one residual electromagnetic singlet contribution with three weak-adjoint contributions under the normalization adopted in this closure rule. It is referred to below as the singlet-plus-adjoint compact

factor. The numerical effect of the second-order term is to refine the leading weak-scale candidate. Its independent derivation and uniqueness remain part of the PR-III precision future research. The corrected hierarchy action is

$$S_{\text{EW}}^{\text{PR,total}} = S_{\text{EW}}^{\text{PR,base}} + \delta S_{\text{EW}}^{(2)}. \quad (121)$$

Numerically,

$$\delta S_{\text{EW}}^{(2)} = -5.892040980909848 \times 10^{-4}, \quad (122)$$

and therefore

$$S_{\text{EW}}^{\text{PR,total}} = 37.17945086706573. \quad (123)$$

The compact-boundary weak displacement anchor is then

$$\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} \exp\left(-S_{\text{EW}}^{\text{PR,total}}\right) = 173.67059834472832 \text{ GeV}. \quad (124)$$

5.5 Generated Weak Order Scale and Fermi Constant

The compact hierarchy now provides a candidate weak scale and the corresponding Fermi constant. These quantities are evaluated from the inherited boundary ledger and fundamental constants, while experimental values are introduced only afterward for diagnostic comparison. Using

$$v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}}, \quad (125)$$

we obtain

$$v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}. \quad (126)$$

The corresponding Fermi constant is

$$G_F^{\text{PR}} = \frac{1}{\sqrt{2} (v_{\text{EW}}^{\text{PR}})^2} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}. \quad (127)$$

These values are generated from the compact-boundary ledger and fundamental constants. They are not inserted as generation inputs. For diagnostic comparison, the reference weak scale implied by the measured Fermi constant is

$$v_{\text{EW}}^{\text{ref}} = 246.2196507941374 \text{ GeV}. \quad (128)$$

The relative residual of the PR-II candidate is

$$\frac{v_{\text{EW}}^{\text{PR}} - v_{\text{EW}}^{\text{ref}}}{v_{\text{EW}}^{\text{ref}}} = -2.23 \times 10^{-7}. \quad (129)$$

Equivalently, the Fermi-constant residual is

$$\frac{G_F^{\text{PR}} - G_F^{\text{ref}}}{G_F^{\text{ref}}} = 4.46 \times 10^{-7}. \quad (130)$$

Table 4: Compact-boundary weak-scale and Fermi-constant closure.

Quantity	PR-II Candidate	Status
$\sin^2 \theta_W^{\text{PR}}$	0.231355763298189	Compact-boundary weak-mixing candidate
Ξ_{EW}	2.0099841245909515	Dimensionless weak-order factor
y_T^{PR}	0.997513275401799	Terminal matter–displacement overlap
$\Lambda_{\text{EW}}^{\text{PR}}$	173.67059834472832 GeV	Compact hierarchy anchor
$v_{\text{EW}}^{\text{PR}}$	246.21959589022453 GeV	Weak order scale candidate
G_F^{PR}	$1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}$	Generated Fermi constant

5.6 Electroweak Mass Geometry

After establishing the electroweak order scale from the compact spectral hierarchy, we now define the resulting mass geometry of the gauge sector. In the Projection Relativity framework, the masses of the W and Z bosons are not independent parameters but are emergent excitations of the broken directions of the parent compact fiber. This section formalizes the structural relationships between the charged and neutral modes, ensuring that the tree-level consistency conditions of the Standard Model—including the masslessness of the residual photon and the preservation of the $\rho = 1$ relation—are recovered as strict consequences of the projection-locking geometry. The electroweak mass geometry follows from the relations

$$M_W^2 = \frac{g^2 (v_{\text{EW}}^{\text{PR}})^2}{4}, \quad (131)$$

and

$$M_Z^2 = \frac{(g^2 + g'^2) (v_{\text{EW}}^{\text{PR}})^2}{4}. \quad (132)$$

The photon remains massless, $M_\gamma = 0$. The tree-level mass relation remains $M_W = M_Z \cos \theta_W$. The tree-level rho parameter is therefore

$$\rho_{\text{EW}} = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (133)$$

The numerical W/Z masses depend on the electromagnetic coupling evaluated at the weak scale. PR-I fixed the low-energy compact electromagnetic boundary,

$$\alpha_{\text{PR,bc}}^{-1} = 137.036361812007. \quad (134)$$

A full electroweak precision comparison requires the electromagnetic running layer

$$\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z). \quad (135)$$

Accordingly, the present section treats the W/Z mass geometry as structurally closed, while precision numerical W/Z comparison is classified as a radiative/running matching task.

5.7 No-Free-Parameter Status

The weak-scale closure uses only the PR-I compact-boundary ledger and fundamental constants:

$$q_{\text{bc}}, \quad c_{\text{bc}}, \quad T_Y, \quad \bar{M}_{\text{Pl}}, \quad \hbar, \quad c, \quad G_N. \quad (136)$$

The following quantities are not used as generation inputs:

$$G_F, \quad v_{EW}, \quad M_W, \quad M_Z, \quad m_t, \quad \Lambda_{EW}. \quad (137)$$

They are used as diagnostic references. The weak-scale closure chain is

$$\boxed{\{q_{bc}, c_{bc}, T_Y\} \rightarrow \sin^2 \theta_W^{\text{PR}} \rightarrow \Xi_{EW} \rightarrow S_{EW}^{\text{PR, total}} \rightarrow \Lambda_{EW}^{\text{PR}} \rightarrow v_{EW}^{\text{PR}} \rightarrow G_F^{\text{PR}}.} \quad (138)$$

The Fermi constant is no longer an independent input at this stage. It is generated from the compact weak-scale hierarchy candidate.

5.8 Falsification Bounds

The derivation of the electroweak scale is constrained by the same geometric rigidity that defines the compact fiber. The hierarchy possesses explicit failure modes that falsify the PR-II architecture if violated. First, the omission of the determinant term,

$$\ln \left(\frac{q_{bc}}{c_{bc}} \right), \quad (139)$$

would fail to correctly weight the compact stiffness relative to the boundary cofactor, thereby failing to reproduce the observed weak hierarchy. Second, ignoring the $3 + 1$ projection-trace orientation factor,

$$\ln \left(\frac{4}{3} \right), \quad (140)$$

would break the integrity of the compact hierarchy action by decoupling the geometric orientation of the non-Abelian fiber. Third, any attempts to generate a dimensionful weak scale from the dimensionless order factor Ξ_{EW} in isolation is invalid; the framework requires a terminal dimensionful reference scale—here, the reduced Planck mass $\bar{M}_{\text{Pl}}$ —to anchor the projection. Finally, any derivation that imports the experimental values of

$$G_F, \quad v_{EW}, \quad M_W, \quad M_Z, \quad m_t, \quad \Lambda_{EW} \quad (141)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The model remains viable only so long as these observables emerge as outputs of the hierarchy closure. With the weak-mixing and electroweak gauge-ledger established, we conclude the derivation of the scale. The subsequent section demonstrates how the compact non-Abelian phase projection recovers the full breadth of charged-current and neutral-current weak interactions, including beta-transition topology, Z -couplings, and the low-energy Fermi contact limit.

6 Compact Order Mode and Tree-Level Scalar Coupling Ledger

Projection locking leaves one unbroken electromagnetic direction and three broken electroweak directions. This section parameterizes the associated order field, separates the compact orientation modes from the remaining scalar excitation, and derives the tree-level scalar couplings that follow from the order-field kinetic term. The previous section generated the weak order scale,

$$\boxed{v_{EW}^{\text{PR}} = 246.21959589022453 \text{ GeV},} \quad (142)$$

and the corresponding Fermi constant,

$$\boxed{G_F^{\text{PR}} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}.} \quad (143)$$

The projection-locking sections also recovered the tree-level gauge-boson mass ledger:

$$M_\gamma = 0, \quad M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v_{\text{EW}}^2}{4}. \quad (144)$$

A complete tree-level electroweak ledger also requires the scalar order-mode associated with the locked direction. This section supplies that tree-level scalar-coupling ledger. It does not claim a full electroweak precision completion. In particular, the physical scalar mass, electroweak radiative corrections, electromagnetic running to the weak scale, and strong-sector loop inputs are deferred to the PR-III precision layer.

6.1 Compact Order Field and Broken-Direction Counting

To systematically map the scalar sector, it is necessary to parameterize the full compact order field around the defined locking direction [11, 12]. We track the internal degrees of freedom as the parent electroweak fiber reduces to the residual electromagnetic boundary. By isolating the radial excitation from the compact orientation modes, we verify the structural counting: exactly one physical scalar degree of freedom survives projection locking, while the broken non-Abelian directions supply the longitudinal components for the massive weak projection modes. The compact electroweak order direction used for projection locking is

$$\Phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} \end{pmatrix}, \quad Y = 1. \quad (145)$$

The full compact order field may be written as

$$\Phi(x) = \frac{1}{\sqrt{2}} U(x) \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix}, \quad (146)$$

where

$$U(x) = \exp \left[\frac{i\pi^a(x) T_a}{v_{\text{EW}}^{\text{PR}}} \right] \quad (147)$$

carries the compact orientation modes, and $h_{\text{PR}}(x)$ is the radial order-mode excitation around the locked electroweak boundary. The parent compact electroweak fiber breaks as

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \quad (148)$$

Thus the number of broken compact directions is

$$\dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (149)$$

The complex order doublet has four real components. Projection locking therefore leaves $4 - 3 = 1$ physical scalar order-mode degree of freedom h_{PR} . The three compact orientation modes become the longitudinal components of the massive weak projection modes:

$$\pi^a \longrightarrow W_L^+, W_L^-, Z_L. \quad (150)$$

The remaining scalar h_{PR} is the tree-level compact order-mode excitation associated with the electroweak locking boundary.

6.2 Gauge-Boson Couplings of the Compact Order Mode

After isolating the single physical scalar order-mode h_{PR} , we map its dynamics relative to the massive weak projection modes. By evaluating the compact order kinetic term in unitary gauge, we show that the same geometric constraints that generate W and Z masses also fix their tree-level scalar coupling ratios. The tree-level scalar gauge-boson couplings follow from the compact order kinetic term,

$$\mathcal{L}_{\text{order}} = (D_\mu \Phi)^\dagger (D^\mu \Phi). \quad (151)$$

In unitary gauge,

$$U(x) = I, \quad (152)$$

so that

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix}. \quad (153)$$

The charged and neutral mass terms become

$$\mathcal{L}_{\text{order}} \supset M_W^2 W_\mu^+ W^{-\mu} \left(1 + \frac{h_{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} \right)^2, \quad (154)$$

and

$$\mathcal{L}_{\text{order}} \supset \frac{1}{2} M_Z^2 Z_\mu Z^\mu \left(1 + \frac{h_{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} \right)^2. \quad (155)$$

Expanding the charged term gives

$$\mathcal{L}_{\text{order}} \supset M_W^2 W_\mu^+ W^{-\mu} + \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}} h_{\text{PR}} W_\mu^+ W^{-\mu} + \frac{M_W^2}{(v_{\text{EW}}^{\text{PR}})^2} h_{\text{PR}}^2 W_\mu^+ W^{-\mu}. \quad (156)$$

The neutral term gives

$$\mathcal{L}_{\text{order}} \supset \frac{1}{2} M_Z^2 Z_\mu Z^\mu + \frac{M_Z^2}{v_{\text{EW}}^{\text{PR}}} h_{\text{PR}} Z_\mu Z^\mu + \frac{M_Z^2}{2(v_{\text{EW}}^{\text{PR}})^2} h_{\text{PR}}^2 Z_\mu Z^\mu. \quad (157)$$

In the conventional vertex normalization, the single-scalar gauge couplings are therefore

$$\boxed{g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}}, \quad g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}}. \quad (158}$$

We can now use the same compact order direction that generates the W/Z masses to also fix the tree-level scalar coupling ratios to those massive weak projection modes.

6.3 Fermion Coupling Ledger

The PR-II flavor sections generate effective matter displacement overlaps rather than inserting fitted Yukawa [17] matrices. At low energy, the scalar coupling is determined by the dependence of the effective fermion mass operator on the scalar order mode. In general,

$$g_{hff}^{\text{PR}} = \left. \frac{\partial m_f^{\text{PR}}(h_{\text{PR}})}{\partial h_{\text{PR}}} \right|_{h_{\text{PR}}=0}. \quad (159)$$

For the tree-level matching adopted in this framework, the effective mass is taken to depend linearly on the order mode:

$$m_f^{\text{PR}}(h_{\text{PR}}) = m_f^{\text{PR}} \left(1 + \frac{h_{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} \right). \quad (160)$$

Under this linear matching condition,

$$\boxed{g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}.} \quad (161)$$

The linear dependence is a tree-level matching assumption; its derivation from the full compact overlap operator is assigned to the PR-III scalar precision future work. We find that the same overlap quantities that generate fermion mass candidates also determine the corresponding tree-level scalar coupling ledger. For the charged-lepton sector, this means

$$\boxed{g_{hee}^{\text{PR}} = \frac{m_e^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}, \quad g_{h\mu\mu}^{\text{PR}} = \frac{m_\mu^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}, \quad g_{h\tau\tau}^{\text{PR}} = \frac{m_\tau^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}.} \quad (162)$$

For the quark sector, the same relation applies after the QCD running and threshold convention has been specified:

$$\boxed{g_{hqq}^{\text{PR}} = \frac{m_q^{\text{PR}}(\mu)}{v_{\text{EW}}^{\text{PR}}}.} \quad (163)$$

For the Majorana-dominant neutrino sector, the leading mass operator is Planck suppressed and quadratic in the weak order scale. Its scalar coupling structure is therefore part of the neutrino precision layer and is treated together with the Majorana boundary rather than as an unsuppressed Dirac Yukawa coupling.

6.4 Tree-Level Longitudinal Weak-Boson Ledger

With the scalar couplings for the W and Z bosons defined, we can trace these longitudinal modes back to the broken compact orientation of the parent fiber. Mapping these modes reveals how gauge mass terms and scalar interactions combine to maintain the tree-level cancellation needed for longitudinal unitarity. Specifically, these broken modes provide the directions for the longitudinal weak bosons:

$$\boxed{\pi^a \rightarrow W_L^\pm, Z_L.} \quad (164)$$

The remaining scalar order mode h_{PR} couples to $W^\pm$ and Z with the tree-level ratios derived above. Accordingly, PR-II contains the same structural ingredients required for the tree-level cancellation pattern [18] of longitudinal weak-boson scattering:

$$\boxed{W_L W_L \quad \text{requires} \quad \{W/Z \text{ mass terms}, h_{\text{PR}} W W, h_{\text{PR}} Z Z\}.} \quad (165)$$

This section does not compute the full high-energy scattering amplitudes. It establishes the tree-level scalar coupling ledger that such an amplitude calculation would require. The amplitude-level unitarity analysis, including radiative corrections, is deferred to PR-III future research.

6.5 Scalar Mass and Compact Hessian Target

While the tree-level couplings between the scalar order-mode and the gauge ledger are fixed by the compact geometry, the physical mass of the scalar itself requires a dynamical evaluation of the vacuum curvature. It is imperative to clarify that PR-II does not claim a zero-new-parameter derivation of the scalar mass

at this stage. Instead of inserting the observed scalar mass as an empirical fit, this framework formally defines the scalar mass as a geometric target, setting the stage for the PR-III precision program.

The scalar/order-mode coupling ledger is structurally recovered at tree level. However, the scalar mass itself is not inserted in as part of the calculation. In a compact-boundary formulation, the scalar mass must arise from the curvature of the order-locking action:

$$\boxed{m_{h,\text{PR}}^2 = \left. \frac{\partial^2 V_{\text{lock}}^{\text{PR}}}{\partial h_{\text{PR}}^2} \right|_{h_{\text{PR}}=0}}. \quad (166)$$

Equivalently, $m_{h,\text{PR}}^2$ is the compact order-mode Hessian evaluated at the locked electroweak boundary. The present manuscript does not claim a zero-new-parameter derivation of the scalar mass. The observed scalar mass and scalar self-couplings are not used as inputs anywhere in the closure chain. They are deferred to the PR-III electroweak precision program, whose tasks include:

$$\boxed{m_{h,\text{PR}}, \quad \lambda_{\text{lock}}^{\text{PR}}, \quad \Delta r_{\text{PR}}, \quad \alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z)}, \quad (167)$$

together with the scalar contribution to longitudinal weak-boson unitarity and loop-level precision observables.

6.6 Tree-Level Electroweak Ledger and PR-III Obligations

The preceding derivations collectively establish the foundational electroweak architecture, demonstrating that the gauge symmetries, mass matrices, physical currents, and interaction vertices emerge as geometric consequences of the compact non-Abelian phase projection. We also demarcate the strict boundaries of the current manuscript by explicitly cataloging the radiative, running, and higher-order precision tasks deferred to the subsequent PR-III program.

The electroweak sector of PR-II is therefore complete at the tree-level gauge, current, mass, and scalar-coupling ledger level:

$$\boxed{SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}}, \quad (168)$$

$$\boxed{M_\gamma = 0, \quad M_W, \quad M_Z, \quad \rho_{\text{EW}} = 1}, \quad (169)$$

$$\boxed{\mathcal{L}_{cc}, \quad \mathcal{L}_Z, \quad \mathcal{L}_F}, \quad (170)$$

and

$$\boxed{g_{hWW}^{\text{PR}}, \quad g_{hZZ}^{\text{PR}}, \quad g_{hff}^{\text{PR}}}. \quad (171)$$

It is not yet complete at the full electroweak precision level. The remaining electroweak precision obligations are explicitly assigned to PR-III:

$$\boxed{\begin{array}{l} \text{scalar mass,} \quad \text{scalar self-couplings,} \quad \text{electromagnetic running,} \\ \text{loop-level weak precision,} \quad \text{strong-sector radiative inputs.} \end{array}} \quad (172)$$

This division keeps the present manuscript focused. PR-II establishes the compact-boundary origin of the tree-level electroweak and flavor ledger. PR-III must complete the radiative, scalar-Hessian, and strong-sector precision layers.

6.7 Negative Controls and Falsification Boundary

The scalar ledger is geometrically rigid and presents several immediate failure modes. First, the order field must intrinsically supply four real components before locking:

$$\Phi \in \mathbb{C}^2 \implies 4 \text{ real components.} \quad (173)$$

If the order field does not meet this dimensional prerequisite, the geometric counting that isolates a single physical scalar order-mode, $4 - 3 = 1$, mechanically fails.

Second, the locked direction must remain neutral under the residual generator,

$$Q\Phi_0 = 0. \quad (174)$$

If $Q\Phi_0 \neq 0$, the unbroken electromagnetic boundary is not preserved, fundamentally breaking the projection.

Third, the scalar gauge couplings must follow from the exact same order kinetic term that generates the massive weak projection modes:

$$g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}}, \quad g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}}. \quad (175)$$

If these coupling ratios require independent tuning or fail to match this derivation, the scalar ledger is rendered mathematically inconsistent with the projection-locking mass ledger.

Fourth, under the linear tree-level matching condition, the Dirac fermion scalar couplings must satisfy

$$g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}. \quad (176)$$

Failure of this relation would invalidate the adopted linear matching condition, while a nonlinear mass dependence would require evaluation through

$$g_{hff}^{\text{PR}} = \left. \frac{\partial m_f^{\text{PR}}(h_{\text{PR}})}{\partial h_{\text{PR}}} \right|_{h_{\text{PR}}=0}. \quad (177)$$

Finally, any attempt to insert the observed scalar mass as an empirical input is prohibited:

$m_h \text{ is not a generation input in this manuscript.}$

(178)

Such an insertion violates the zero-new-parameter discipline of PR-II. The scalar mass must be derived dynamically from the compact order-Hessian closure. Until that derivation is complete, it remains a PR-III precision target. With the compact order-mode and tree-level scalar coupling ledger established, we conclude this sector of the projection architecture. The subsequent section utilizes the compact non-Abelian current structure to recover the full breadth of charged-current and neutral-current weak interactions.

7 Charged and Neutral Current Recovery

The weak interactions are the physical currents induced by state-motion along the $SU(2)_L$ fiber [19]. We now demonstrate that the charged-current and neutral-current Lagrangians are the low-energy contact limits of the parent projection geometry. The central chain is defined as

$SU(2)_L \rightarrow \{T^\pm, T_3\} \rightarrow \{W^\pm, Z, A\} \rightarrow \{\mathcal{L}_{cc}, \mathcal{L}_Z, \mathcal{L}_F\}.$

(179)

Within this map, the photon is preserved as the massless residual Q -projection, while the $W^\pm$ and Z manifest as the massive excitation modes of the broken electroweak directions.

7.1 Charged-Current Topological Ladder

The transition from static geometry to active weak processes begins with the definition of the topological ladder. The "charged-current" interactions such as the decay of a neutron or the transition of a quark—are not governed by extrinsic force rules. Rather, they are the intrinsic response of the $SU(2)_L$ fiber as a state undergoes a discrete geometric shift. We formalize these transitions as simple "ladder moves" within the internal manifold, demonstrating that the standard beta-decay topology is the mechanical result of state-motion across the compact-boundary projection. The charged-current ladder operators are

$$T^\pm = T_1 \pm iT_2. \quad (180)$$

In the weak doublet basis,

$$|\text{up}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\text{down}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (181)$$

they satisfy

$$\boxed{T^+|\text{down}\rangle = |\text{up}\rangle, \quad T^-|\text{up}\rangle = |\text{down}\rangle.} \quad (182)$$

The weak charged-current transitions are compact-isospin ladder moves within a weak doublet. The residual electromagnetic generator is

$$\boxed{Q = T_3 + \frac{Y}{2}.} \quad (183)$$

The ladder operators carry one unit of residual electromagnetic charge:

$$[Q, T^+] = +T^+, \quad [Q, T^-] = -T^-. \quad (184)$$

Accordingly, T^+ is the charge-raising compact transition and T^- is the charge-lowering compact transition. For the quark doublet,

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}, \quad (185)$$

and for the lepton doublet,

$$Q_\nu = 0, \quad Q_e = -1. \quad (186)$$

The ordinary beta-transition topology is therefore

$$\boxed{d \rightarrow u + W^-}, \quad (187)$$

followed by

$$\boxed{W^- \rightarrow e^- + \bar{\nu}_e.} \quad (188)$$

Charge conservation is immediate:

$$-\frac{1}{3} = \frac{2}{3} - 1, \quad -1 = -1 + 0. \quad (189)$$

In PR-II language, this transition is a compact-boundary ladder move:

$$\boxed{T^\pm \rightarrow W^\pm \rightarrow \text{charged-current weak transition}.} \quad (190)$$

7.2 Charged Weak Projection Modes and the Fermi Limit

The $W^\pm$ bosons are the specific massive projection modes associated with the non-Abelian rotations of the parent fiber. By evaluating these modes in the low-momentum limit ($q^2 \ll M_W^2$), we demonstrate that the standard four-current Fermi interaction is recovered as a geometric residue [15]. This section concludes with the precision numerical generation of the Fermi constant (G_F), establishing the strength of the weak interaction as a deterministic consequence of the compact spectral hierarchy. The charged weak projection modes are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} \left(W_\mu^1 \mp i W_\mu^2 \right). \quad (191)$$

A convenient Fermi-normalized charged-current convention is

$$\mathcal{L}_{\text{cc}} = -\frac{g}{2\sqrt{2}} \left(J_{F,\mu}^+ W^{-\mu} + J_{F,\mu}^- W^{+\mu} \right). \quad (192)$$

Here $J_F^\pm$ are normalized so that integrating out the $W^\pm$ gives the standard Fermi coefficient. Equivalently, if one defines the compact projection current directly as $J_{\text{PR}}^{\pm\mu} = \bar{\psi}_L \gamma^\mu T^\pm \psi_L$, then conventional factors of 2 are absorbed into the current normalization. The invariant matching relation is independent of this convention. At momentum transfer much smaller than the charged weak mass, $q^2 \ll M_W^2$, the massive charged weak projection mode can be integrated out. The result is the four-current Fermi contact interaction

$$\boxed{\mathcal{L}_F = -\frac{G_F}{\sqrt{2}} J_{F,\mu}^+ J_F^{-\mu}.} \quad (193)$$

The coefficient satisfies

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (194)$$

Using

$$M_W^2 = \frac{g^2 (v_{\text{EW}}^{\text{PR}})^2}{4}, \quad (195)$$

this becomes

$$\boxed{\frac{G_F}{\sqrt{2}} = \frac{1}{2(v_{\text{EW}}^{\text{PR}})^2}.} \quad (196)$$

Thus the low-energy Fermi interaction is the contact limit of the massive charged projection modes. Using the weak-scale candidate from Section 5.5,

$$\boxed{v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV},} \quad (197)$$

the generated Fermi constant is

$$\boxed{G_F^{\text{PR}} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}.} \quad (198)$$

7.3 Neutral Current Projection

While the charge-current ladder handles identity shifts, the neutral sector describes the state response orthogonal to the electromagnetic boundary [2]. The physical photon and the Z boson are not independent entities but are the resulting eigenstates of a single geometric rotation within the parent fiber. By formalizing the neutral-current projection, we demonstrate that the fermion coupling assignments emerge as the trigonometric consequences of the manifold's orientation. The Z -mediated interaction remains locked to

the same compact spectral scale as the charged sector, ensuring that the entire neutral ledger is a consistent hardware output of the projection-locking mechanism. The neutral modes are

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu. \quad (199)$$

The photon A_μ couples to the residual electromagnetic generator Q . The Z_μ couples to the orthogonal neutral-current projection. For a fermion f , the left- and right-handed Z -projection charges are

$$g_L^f = T_3^f - Q_f \sin^2 \theta_W, \quad g_R^f = -Q_f \sin^2 \theta_W. \quad (200)$$

Equivalently, in the vector/axial convention,

$$g_V^f = g_L^f + g_R^f = T_3^f - 2Q_f \sin^2 \theta_W, \quad (201)$$

and

$$g_A^f = g_L^f - g_R^f = T_3^f. \quad (202)$$

The neutral-current interaction may be written as

$$\mathcal{L}_Z = -\frac{g}{2 \cos \theta_W} Z_\mu \bar{f} \gamma^\mu (g_V^f - g_A^f \gamma^5) f. \quad (203)$$

In chiral form, this is equivalently

$$\mathcal{L}_Z = -\frac{g}{\cos \theta_W} Z_\mu \bar{f} \gamma^\mu (g_L^f P_L + g_R^f P_R) f. \quad (204)$$

The neutral projection coupling is

$$g_Z = \frac{g}{\cos \theta_W} = \sqrt{g^2 + g'^2}. \quad (205)$$

The neutral mass relation is

$$M_Z^2 = \frac{g_Z^2 (v_{\text{EW}}^{\text{PR}})^2}{4}. \quad (206)$$

At low momentum transfer, $q^2 \ll M_Z^2$, integrating out the massive neutral projection mode gives the neutral-current contact coefficient

$$\frac{g_Z^2}{8M_Z^2} = \frac{1}{2(v_{\text{EW}}^{\text{PR}})^2} = \frac{G_F^{\text{PR}}}{\sqrt{2}}. \quad (207)$$

7.4 Tree-Level Neutral-Current Couplings

Using the compact-boundary weak-mixing candidate

$$\sin^2 \theta_W^{\text{PR}} = 0.231355763298189, \quad (208)$$

the tree-level neutral-current couplings are shown in Table 5. This Table uses the standard tree-level neutral-current definitions,

$$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W^{\text{PR}}, \quad g_A^f = T_3^f. \quad (209)$$

While a precision audit against experimental neutral-current data involves standard radiative matching and scale-dependent running, the current ledger remains focused on the tree-level foundation. The architectural claim is that the non-Abelian projection recovers the structural coupling pattern of the Standard Model as a geometric necessity of the locked parent fiber [20].

Table 5: Tree-level neutral-current couplings from the compact-boundary weak-mixing value.

Fermion	Q_f	T_3^f	g_V^f	g_A^f
ν_e	0	+1/2	+0.500000000	+0.500000000
e	-1	-1/2	-0.037288473	-0.500000000
u	+2/3	+1/2	+0.191525649	+0.500000000
d	-1/3	-1/2	-0.345762824	-0.500000000

7.5 Mass Geometry and Diagnostic Weak-Scale Coupling

The final structural check of the gauge sector involves the mass values of the W and Z bosons. While the ratios and symmetries of these masses are now geometrically locked, their specific numerical values are scale-dependent, governed by the running of the electromagnetic coupling from the PR-I low-energy boundary ($\alpha \approx 1/137$) to the weak scale ($\alpha(M_Z) \approx 1/128$). A diagnostic tree-level comparison is performed using standard running-coupling values to show that the framework naturally lands within the physical regime. This sets the stage for the final "Running Layer" of the theory, where the transition from the macroscopic boundary to the microscopic scale is derived rather than matched. The PR-II mass geometry is closed by

$$M_W = M_Z \cos \theta_W, \quad \rho_{EW} = 1. \quad (210)$$

Numerical weak-boson masses also require the electromagnetic coupling evaluated at the weak scale:

$$e^2(M_Z) = 4\pi\alpha(M_Z). \quad (211)$$

PR-I fixed the low-energy electromagnetic boundary, $\alpha_{\text{PR},bc}^{-1} = 137.036361812007$, but the running map $\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z)$ is a radiative matching layer. Therefore, precision W/Z mass comparison is not claimed in this section. As a diagnostic tree-level branch, using a weak-scale electromagnetic coupling near

$$\boxed{\alpha(M_Z)^{-1} = 127.955} \quad (212)$$

with the PR weak-mixing and weak-scale candidates gives

$$\boxed{M_W^{\text{PR}} \simeq 80.210045 \text{ GeV}, \quad M_Z^{\text{PR}} \simeq 91.488408 \text{ GeV}.} \quad (213)$$

These values are useful for current-recovery bookkeeping, but the zero-new-parameter precision comparison requires the electromagnetic running layer to be derived rather than inserted.

7.6 Interpretation

The charged-current and neutral-current structures recover the expected tree-level weak interaction ledger from compact non-Abelian projection geometry $T^\pm \rightarrow W^\pm \rightarrow \mathcal{L}_{cc} \rightarrow \mathcal{L}_F$, and $T_3, Q \rightarrow Z_\mu \rightarrow \mathcal{L}_Z$.

Weak decay and neutral-current scattering are not separate additions to PR-II. They are the current structures induced by compact electroweak projection locking. In this language, $W^\pm$ and Z are not primitive independent inputs. They are massive projection excitation modes of the broken electroweak fiber. The photon remains massless because it is the unbroken residual electromagnetic projection mode $M_\gamma = 0$.

7.7 Falsification Bounds

The recovery of the physical weak currents is a deterministic output of state-motion along the gauge fiber. The architecture is falsified if it cannot preserve the following contact limits and charge topologies.

First, the charged-current coefficient must satisfy the matching condition

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (214)$$

Using an alternative normalization such as $g^2/4M_W^2$ would fail to recover the physical Fermi interaction, rendering the solution phenomenologically invalid.

Second, the ladder operators must retain their non-Abelian imaginary structure,

$$T^\pm = T_1 \pm iT_2. \quad (215)$$

Removing the factor of i would destroy the charged-current topological ladder, effectively collapsing the non-Abelian structure into an Abelian one.

Third, the beta-transition bookkeeping must correctly preserve residual charge conservation,

$$d \rightarrow u + W^-. \quad (216)$$

Assigning an incorrect charge to the W excitation would explicitly violate the residual $U(1)_{\text{em}}$ symmetry.

Fourth, the vector coupling must preserve the factor of 2 defining the chiral-neutral projection:

$$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W. \quad (217)$$

Omitting this factor would yield incorrect neutral-current couplings, failing to match the established Z -boson decay data. Finally, the right-handed neutral-current coupling must maintain the sign

$$g_R^f = -Q_f \sin^2 \theta_W. \quad (218)$$

A sign inversion here would break the tree-level Z -coupling ledger, misrepresenting the chiral interaction of the fermion sector.

The current recovery step is falsifiable. To remain viable, the PR-II framework must reproduce the charged-current ladder, beta-transition charge conservation, the Fermi coefficient, the neutral-current Z -couplings, and the tree-level contact matching conditions. The failure of any single identity would break the compact non-Abelian current-recovery program. Now that we have established the gauge-current sector, the subsequent section completes the PR-II architecture by constructing the fermion-representation ledger. This analysis verifies that the compact electroweak fiber admits a mathematically consistent, anomaly-free chiral matter assignment, ensuring the framework is robust against internal topological contradictions.

8 Fermion Representation Ledger and Anomaly Cancellation

The recovery of the charged and neutral current structures in the preceding sections establishes the dynamical viability of the compact electroweak projection. However, for a microscopic extension to be physically consistent, it must support a chiral fermion representation that is free of both local and global gauge anomalies. This section verifies that the parent compact fiber,

$$SU(2)_L \times U(1)_Y, \quad (219)$$

operating under the residual electromagnetic generator,

$$Q = T_3 + \frac{Y}{2}, \quad (220)$$

admits the minimal matter ledger required to populate the electroweak and color sectors. This analysis is structural, defining the internal configuration of a single generation. While the derivation of the three-generation count is addressed through the compact flavor-sheet geometry in Section 9, the present section establishes the anomaly-free criteria that any projected generation must satisfy to maintain the manifold's integrity.

8.1 Left-Handed Weyl Convention

All fermions are written as left-handed Weyl fields [21]. Accordingly, the ordinary right-handed fermions are represented by their left-handed conjugates:

$$u_R \rightarrow u_R^c, \quad d_R \rightarrow d_R^c, \quad e_R \rightarrow e_R^c. \quad (221)$$

This convention is essential for anomaly bookkeeping. The conjugate fields transform in conjugate gauge representations and carry opposite hypercharge relative to the corresponding physical right-handed fields. For one generation, the electroweak–color representation ledger is

$$Q_L : (3, 2)_{1/3}, \quad u_R^c : (\bar{3}, 1)_{-4/3}, \quad d_R^c : (\bar{3}, 1)_{2/3}, \quad (222)$$

and

$$L_L : (1, 2)_{-1}, \quad e_R^c : (1, 1)_2. \quad (223)$$

The subscript denotes hypercharge Y , using the residual electromagnetic generator 28. An optional sterile neutrino singlet may be added, $\nu_R^c : (1, 1)_0$. Because it is neutral under $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$, this sterile singlet does not affect gauge-anomaly cancellation.

8.2 Residual Electromagnetic Charges

To verify the internal consistency of the matter sector, we map the specific charge residues for individual fermion states following the projection locking step. The electric charge is not a fundamental label, but the specific geometric coordinate a state occupies on the residual electromagnetic boundary. By applying the locking relation $Q = T_3 + Y/2$ to the chiral Weyl multiplets, we demonstrate that the observed fractional and integer charges of the Standard Model are the deterministic outputs of the manifold’s orientation. We show that while the parent weak representation is chiral, the resulting electromagnetic projection is vector-like, providing the necessary symmetry for the stability of the long-range force sector. The residual electromagnetic charges follow the residual electromagnetic generator 28 For the left-handed quark doublet,

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad Y = \frac{1}{3}, \quad (224)$$

we obtain

$$Q_u = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}, \quad Q_d = -\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}. \quad (225)$$

For the left-handed lepton doublet,

$$L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Y = -1, \quad (226)$$

we obtain

$$Q_\nu = \frac{1}{2} - \frac{1}{2} = 0, \quad Q_e = -\frac{1}{2} - \frac{1}{2} = -1. \quad (227)$$

The conjugate singlets carry

$$Q(u_R^c) = -\frac{2}{3}, \quad Q(d_R^c) = +\frac{1}{3}, \quad Q(e_R^c) = +1. \quad (228)$$

Conjugating back gives the ordinary right-handed charges,

$$Q(u_R) = +\frac{2}{3}, \quad Q(d_R) = -\frac{1}{3}, \quad Q(e_R) = -1. \quad (229)$$

The residual electromagnetic boundary is vectorlike after projection locking, even though the parent weak representation is chiral.

Table 6: One-generation left-handed Weyl fermion ledger.

Field	$SU(3)_c$	$SU(2)_L$	Y
$Q_L = (u_L, d_L)$	3	2	1/3
u_R^c	$\bar{3}$	1	-4/3
d_R^c	$\bar{3}$	1	2/3
$L_L = (\nu_L, e_L)$	1	2	-1
e_R^c	1	1	2
ν_R^c optional	1	1	0

8.3 Pure Color Anomaly

The pure color anomaly, represented by the $[SU(3)_c]^3$ triangle diagram, identifies potential points of mathematical divergence within the chiral representation. The cancellation of this anomaly is a validation of the manifold's hardware integrity. We evaluate the color-charge multiplicities across the projected states, demonstrating that the representation is vectorlike with respect to the strong sector. By recovering a null coefficient, we confirm that the color currents are conserved and that the manifold is free of color-leakage at the compact boundary. Using the convention that a color triplet contributes +1, an anti-triplet contributes -1, and the quark doublet has two weak components, the one-generation coefficient is

$$\boxed{[SU(3)_c]^3 : \quad 2 - 1 - 1 = 0.} \quad (230)$$

The color representation is vectorlike in the left-handed Weyl ledger, and the pure color anomaly cancels [8].

8.4 Mixed Color–Hypercharge Anomaly

Ensuring the integrity of the manifold requires more than auditing individual gauge groups; we must also verify the compatibility of their cross-couplings. The mixed color–hypercharge anomaly, denoted by $[SU(3)_c]^2 U(1)_Y$, measures the dynamical interference between the strong-sector color current and the hypercharge phase fiber. This cancellation is not a coincidental alignment but a requirement for the co-existence of color and electromagnetic projections. Evaluating the weighted sum of hypercharges across the color-triplet states, we demonstrate that the compact color ledger is compatible with the hypercharge assignment established in PR-I. The Dynkin index of the fundamental and anti-fundamental color representations is

$$T(3) = T(\bar{3}) = \frac{1}{2}. \quad (231)$$

The one-generation coefficient is

$$\boxed{[SU(3)_c]^2 U(1)_Y : \quad 2 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right) \left(-\frac{4}{3}\right) + \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) = 0.} \quad (232)$$

Therefore the compact color ledger is compatible with the hypercharge assignment.

8.5 Mixed Weak–Hypercharge Anomaly

We check the internal electroweak consistency by the interference between the non-Abelian weak fiber and the hypercharge phase. The mixed weak–hypercharge anomaly, $[SU(2)_L]^2 U(1)_Y$, represents the balancing act between the baryonic and leptonic sectors of the matter ledger. We audit the contribution of the weak

doublets, demonstrating that the quark and lepton multiplicities are calibrated to cancel the anomaly coefficient. This result confirms that the manifold's hardware is grounded, such that the electroweak currents can propagate across the fiber without mathematical divergence. Where only weak doublets contribute and the one-generation coefficient is

$$\boxed{[SU(2)_L]^2 U(1)_Y : \quad 3 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right) (-1) = 0.} \quad (233)$$

The factor of 3 is the color multiplicity of the quark doublet. Thus the quark and lepton doublets cancel in the mixed weak-hypercharge anomaly.

8.6 Mixed Gravitational-Hypercharge Anomaly

Global compatibility requires the alignment between the microscopic matter ledger and the macroscopic gravitational manifold established in PR-I. The mixed gravitational-hypercharge anomaly, $[\text{grav}]^2 U(1)_Y$, identifies whether the identity-labels of the fermion states induce a global "shear" on the background curvature. This cancellation ensures that the discrete microscopic charges do not destabilize the smooth gravitational response of the projection. We now audit the total hypercharge sum across all projected states, demonstrating that the ledger is balanced against the gravitational foundation inherited from the macroscopic theory. It is proportional to the sum of hypercharges over all left-handed Weyl fields, including multiplicities. For one generation,

$$\boxed{[\text{grav}]^2 U(1)_Y : \quad 6 \left(\frac{1}{3}\right) + 3 \left(-\frac{4}{3}\right) + 3 \left(\frac{2}{3}\right) + 2(-1) + 2 = 0.} \quad (234)$$

Therefore the hypercharge current is free of the mixed gravitational anomaly.

8.7 Cubic Hypercharge Anomaly

A sensitivity test of internal $U(1)_Y$ self-consistency is the cubic hypercharge anomaly, $[U(1)_Y]^3$. This audit measures the volumetric interference of the Abelian phase fiber with itself. Within the PR-II architecture, the specific hypercharge assignments are not arbitrary fit-parameters, but the geometric coordinates required for the manifold to lock. This verifies that these coordinates produce a null cubic coefficient. A zero-result here provides the ultimate validation that the Abelian sector is stable and free of internal phase-divergence, completing the grounding check of the one-generation matter ledger. The one-generation coefficient is

$$\boxed{[U(1)_Y]^3 : \quad 6 \left(\frac{1}{3}\right)^3 + 3 \left(-\frac{4}{3}\right)^3 + 3 \left(\frac{2}{3}\right)^3 + 2(-1)^3 + 2^3 = 0.} \quad (235)$$

Thus the Abelian hypercharge sector is anomaly free for the same representation ledger.

The remaining consistency condition is the global $SU(2)$ anomaly. A chiral $SU(2)$ gauge theory is inconsistent if the number of left-handed weak doublets is odd [9]. For one generation, the number of $SU(2)_L$ doublets is $N_2 = 3 + 1 = 4$. The three quark doublets arise from color multiplicity, and the final doublet is the lepton doublet. Since

$$\boxed{N_2 = 4 \in 2\mathbb{Z},} \quad (236)$$

the Witten $SU(2)$ global anomaly is absent.

8.8 Anomaly-Free One-Generation Ledger

By applying the individual audits of the color, weak, and hypercharge sectors, we now synthesize these findings into a unified, stable blueprint for the matter sector. The Anomaly-Free One-Generation Ledger, representing the set of fermion coordinates required to populate the non-Abelian manifold without inducing mathematical divergence or "current leakage." This specific representation is not merely an empirical list of particles, but the solution compatible with the $SU(2)_L \times U(1)_Y \rightarrow U(1)_{em}$ locking geometry. By satisfying both the local gauge sums and the global Witten condition, this ledger establishes the "Locked-In" generation block upon which the multi-generational architecture of Section 9 is constructed. Combining the local and global checks, the one-generation ledger

$$\mathcal{R}_{EW}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2 \quad (237)$$

is anomaly free:

$$[SU(3)_c]^3 = [SU(3)_c]^2 U(1)_Y = [SU(2)_L]^2 U(1)_Y = [U(1)_Y]^3 = [\text{grav}]^2 U(1)_Y = 0. \quad (238)$$

The Witten global condition is also satisfied:

$$N_2 \in 2\mathbb{Z}. \quad (239)$$

The optional sterile singlet

$$(1, 1)_0 \quad (240)$$

can be included without changing any anomaly coefficient. This anomaly-free ledger is the chiral matter representation compatible with compact non-Abelian phase projection and residual electromagnetic projection locking:

$$\boxed{SU(2)_L \times U(1)_Y \rightarrow U(1)_{em}, \quad Q = T_3 + \frac{Y}{2}.} \quad (241)$$

8.9 Representation Consistency and Anomaly Constraints

The anomaly-cancellation step is not a numerical fit. It is a representation consistency test. The compact electroweak fiber may generate a weak mixing angle, weak scale, and flavor hierarchy only if the matter ledger coupled to it is anomaly free. The result of this section is therefore a structural requirement:

$$\boxed{\mathcal{R}_{EW}^{(1)} \text{ is the minimal anomaly-free chiral ledger compatible with PR-II electroweak projection.}} \quad (242)$$

This ledger will be replicated across compact generation sheets in the next section. Anomaly cancellation by itself does not determine the number of generations, because any integer number of complete anomaly-free copies remains locally anomaly free. The generation-count problem therefore requires an additional compact projection argument, supplied by the broken electroweak projection-sheet structure.

8.10 Falsification Bounds

If the electroweak projection cannot intrinsically carry the specific chiral assignments established below, the entire microscopic extension is mathematically rejected.

First, if right-handed fields are introduced without formally converting them to their left-handed Weyl conjugates, the geometric hypercharge signs invert. This forces the anomaly sums to fail, explicitly breaking the global consistency of the representation.

Second, the lepton doublet cannot be arbitrarily omitted. Its absence leaves the mixed weak–hypercharge anomaly unbalanced, yielding a non-zero residual:

$$3 \left(\frac{1}{2} \right) \left(\frac{1}{3} \right) \neq 0. \quad (243)$$

So that the inclusion of the lepton doublet is a structural requirement to exactly cancel the quark contribution.

Third, the lepton doublet hypercharge is constrained. If it deviates from the exact assignment

$$Y_L^\ell = -1, \quad (244)$$

both the residual electromagnetic charges and the mixed weak–hypercharge anomaly fail to recover the physical matter ledger.

Fourth, the color multiplicity of the quark doublet is structurally indispensable; omitting this $SU(3)_c$ dimensionality explicitly destroys the anomaly cancellation across the mixed gauge boundaries.

Finally, treating hypercharge as synonymous with electric charge, rather than applying the strict projection-locking generator

$$Q = T_3 + \frac{Y}{2}, \quad (245)$$

yields an incorrect residual electromagnetic ledger and breaks the unbroken projection sequence. The falsification boundary for the matter sector is unequivocal:

PR-II must support an anomaly-free chiral fermion ledger.

(246)

If the compact electroweak projection cannot intrinsically carry the representation ledger established above, the entire microscopic extension mathematically fails.

Now with the fermion representation and anomaly-cancellation established, the subsequent section utilizes the compact projection-locking geometry to derive the three-generation ledger and the corresponding generation-sheet flavor hierarchy.

9 Generation Origin and Compact Flavor Sheets

The anomaly-cancellation section has established that one chiral fermion generation is compatible with the compact electroweak projection fiber. However, anomaly cancellation alone does not determine the number of generations. Any integer number of complete anomaly-free copies remains locally anomaly free. The generation problem therefore requires an additional geometric input [22, 23]. In PR-II, the generation ledger is tied to the projection-locking geometry itself. The parent compact electroweak fiber has four compact directions,

$$\dim(SU(2)_L \times U(1)_Y) = 3 + 1 = 4. \quad (247)$$

Projection locking leaves one residual electromagnetic direction,

$$\dim(U(1)_{\text{em}}) = 1. \quad (248)$$

The broken projection space therefore has dimension

$$\dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 4 - 1 = 3. \quad (249)$$

PR-II identifies these three broken compact directions as the three terminal electroweak projection sheets available to the replicated matter ledger:

$$\boxed{N_{\text{sheet}}^{\text{EW}} = 3.} \quad (250)$$

The generation-origin candidate is therefore

$$\boxed{N_{\text{gen}}^{\text{PR}} = N_{\text{sheet}}^{\text{EW}} = \dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3.} \quad (251)$$

This result should be read as a compact projection-origin candidate, not as a consequence of anomaly cancellation alone. Anomaly cancellation verifies that the replicated ledgers are consistent. The projection quotient supplies the geometric reason that the replicated ledger has three compact sheets.

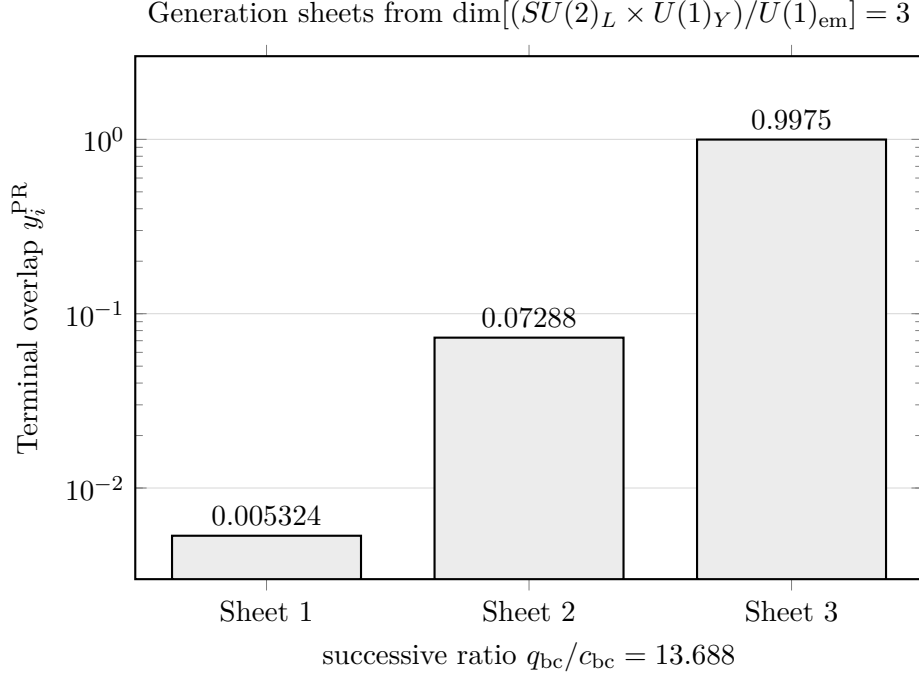


Figure 4: Three compact generation sheets arise from the broken electroweak projection quotient. The terminal matter-displacement overlaps are hierarchically ordered by the determinant ratio $r_G = c_{\text{bc}}/q_{\text{bc}}$.

9.1 Three-Sheet Representation Ledger

In this section, the goal is to replicate the minimal one-generation chiral ledger across the three broken compact projection directions. Because local anomalies are additive and the Witten global anomaly depends on weak-doublet parity, this direct mapping ensures that the multi-sheet architecture inherently preserves the anomaly-free consistency of the parent fiber. The one-generation anomaly-free ledger was

$$\mathcal{R}_{\text{EW}}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2. \quad (252)$$

The three-sheet PR-II ledger is

$$\mathcal{R}_{\text{EW}}^{(3)} = \bigoplus_{i=1}^3 \mathcal{R}_{\text{EW},i}^{(1)}. \quad (253)$$

Since each sheet carries one anomaly-free ledger, local anomalies remain zero:

$$\boxed{\mathcal{A}_{\text{local}}(\mathcal{R}_{\text{EW}}^{(3)}) = 3 \mathcal{A}_{\text{local}}(\mathcal{R}_{\text{EW}}^{(1)}) = 0.} \quad (254)$$

Explicitly,

$$\boxed{[SU(3)_c]^3 = [SU(3)_c]^2 U(1)_Y = [SU(2)_L]^2 U(1)_Y = [U(1)_Y]^3 = [\text{grav}]^2 U(1)_Y = 0.} \quad (255)$$

The Witten $SU(2)$ global anomaly also remains absent. Each generation contains four left-handed weak doublets: three color copies of the quark doublet and one lepton doublet. For three generations, $N_2 = 3(3 + 1) = 12$. Since $12 \in 2\mathbb{Z}$, the global $SU(2)$ anomaly vanishes. The compact electroweak projection admits a three-sheet replicated matter ledger:

$$\boxed{N_{\text{gen}}^{\text{PR}} = 3 \implies \mathcal{R}_{\text{EW}}^{(3)} \text{ is anomaly consistent.}} \quad (256)$$

9.2 Generation-Sheet Overlap Hierarchy

The compact electroweak quotient has three broken directions,

$$\dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (257)$$

PR-II proposes identifying these directions with three fermion-generation sheets. This geometric identification supplies the value $N_{\text{gen}}^{\text{PR}} = 3$, while anomaly cancellation confirms that three replicated chiral ledgers remain consistent. The identification is a closure candidate; it does not follow from coset dimension or anomaly cancellation alone. Given this three-sheet assignment, the compact determinant ratio $r_G = c_{\text{bc}}/q_{\text{bc}}$ is used to define an ordered overlap hierarchy. The hierarchy is generated by the determinant suppression ratio inherited from PR-I:

$$\boxed{r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}} = 0.073056757459128.} \quad (258)$$

Since

$$\boxed{0 < r_G < 1,} \quad (259)$$

this ratio defines a natural compact-boundary suppression factor. For $N_{\text{gen}}^{\text{PR}} = 3$, define the generation-sheet overlap factors

$$\boxed{\eta_i = r_G^{N_{\text{gen}} - i}, \quad i = 1, 2, 3.} \quad (260)$$

Numerically we find that,

$$\boxed{\eta_1 = r_G^2 = 0.005337289810442,} \quad (261)$$

$$\boxed{\eta_2 = r_G = 0.073056757459128,} \quad (262)$$

and

$$\boxed{\eta_3 = 1.} \quad (263)$$

The generation hierarchy is therefore

$$\boxed{\eta_1 < \eta_2 < \eta_3.} \quad (264)$$

The successive hierarchy ratio is fixed:

$$\boxed{\frac{\eta_{i+1}}{\eta_i} = \frac{q_{\text{bc}}}{c_{\text{bc}}} = 13.6879877341867.} \quad (265)$$

The terminal weak matter-displacement overlap found in the weak-scale bridge is

$$\boxed{y_T^{\text{PR}} = 0.997513275401799.} \quad (266)$$

The terminal matter–displacement overlaps on the three generation sheets are

$$y_i^{\text{PR}} = y_T^{\text{PR}} \eta_i. \quad (267)$$

Whereas we solve for,

$$\boxed{y_1^{\text{PR}} = 0.005324017440582, \quad y_2^{\text{PR}} = 0.072875085423290, \quad y_3^{\text{PR}} = 0.997513275401799.} \quad (268)$$

Resulting in,

$$\boxed{y_1^{\text{PR}} < y_2^{\text{PR}} < y_3^{\text{PR}}.} \quad (269)$$

This establishes the first PR-II flavor hierarchy: the three compact electroweak projection sheets naturally carry ordered matter–displacement overlaps. Observed fermion masses are not used to obtain this hierarchy.

9.3 Sector-Specific Compact Charge Norms

The universal generation-sheet hierarchy must be refined by sector-specific compact representation data. For a fermion sector f , define the charge norm

$$\mathcal{C}_f = C_2^{SU(3)}(f) + C_2^{SU(2)}(f_L) + \left(\frac{Y_L}{2}\right)^2 + \left(\frac{Y_R}{2}\right)^2. \quad (270)$$

Here $C_2^{SU(3)}(f)$ is the color quadratic Casimir, $C_2^{SU(2)}(f_L) = 3/4$ for left-handed weak doublets, and Y_L, Y_R are the physical left- and right-handed hypercharges of the sector.

The sector factor is

$$\kappa_f = 1 + r_G \mathcal{C}_f. \quad (271)$$

The sector-specific diagonal overlap operator is then

$$D_f = \kappa_f \text{diag} \left(y_1^{\text{PR}}, y_2^{\text{PR}}, y_3^{\text{PR}} \right). \quad (272)$$

For the charged-lepton sector,

$$\boxed{T_{3,L} = -\frac{1}{2}, \quad Y_L = -1, \quad Y_R = -2, \quad Q = -1.} \quad (273)$$

The compact norm and sector factor are

$$\boxed{\mathcal{C}_\ell = 2, \quad \kappa_\ell = 1.146113514918256.} \quad (274)$$

For the up-quark sector,

$$\boxed{T_{3,L} = +\frac{1}{2}, \quad Y_L = \frac{1}{3}, \quad Y_R = \frac{4}{3}, \quad Q = \frac{2}{3}.} \quad (275)$$

Using $C_2^{SU(3)} = 4/3$ and $C_2^{SU(2)} = 3/4$, the compact norm and factor are

$$\boxed{\mathcal{C}_u = \frac{23}{9}, \quad \kappa_u = 1.186700602395549.} \quad (276)$$

For the down-quark sector,

$$\boxed{T_{3,L} = -\frac{1}{2}, \quad Y_L = \frac{1}{3}, \quad Y_R = -\frac{2}{3}, \quad Q = -\frac{1}{3}.} \quad (277)$$

The compact norm and factor are

$$\boxed{\mathcal{C}_d = \frac{20}{9}, \quad \kappa_d = 1.162348349909173.} \quad (278)$$

For an optional Dirac-neutrino sector,

$$\boxed{T_{3,L} = +\frac{1}{2}, \quad Y_L = -1, \quad Y_R = 0, \quad Q = 0.} \quad (279)$$

The compact norm and factor are

$$\boxed{\mathcal{C}_\nu = 1, \quad \kappa_\nu = 1.073056757459128.} \quad (280)$$

9.4 Sector-Specific Overlap Spectra

The resulting sector-specific overlap spectra take the form of diagonal matrices,

$$D_f = \text{diag}((D_f)_1, (D_f)_2, (D_f)_3), \quad (281)$$

where $f \in \{\ell, u, d, \nu\}$. The complete structural overlap ledger is consolidated in Table 7. The same generation hierarchy ratio appears in every sector:

$$\boxed{\frac{(D_f)_{i+1}}{(D_f)_i} = \frac{q_{bc}}{c_{bc}} = 13.6879877341867.} \quad (282)$$

The generation-sheet hierarchy is universal, while the overall sector normalization is shifted by compact representation data. These spectra are not yet mass formulas. They are compact flavor-overlap ledgers. Physical fermion masses require sector-specific scale rules, running, and projection matching, developed in later sections.

Table 7: Sector-specific overlap spectra. These are structural overlap ledgers, not physical fermion mass predictions.

Sector	Matrix	$(D_f)_1$	$(D_f)_2$	$(D_f)_3$
Charged leptons	D_ℓ	0.006101928	0.083523120	1.143263446
Up quarks	D_u	0.006318015	0.086480908	1.183749605
Down quarks	D_d	0.006188363	0.084706235	1.159457910
Dirac neutrino	D_ν	0.005712973	0.078199103	1.070388361

9.5 Hypercharge-Neutral Order Terms

The sector-specific overlap operators must be compatible with the compact electroweak charge ledger. Using the electroweak order direction with $Y_H = +1$ and its conjugate $\tilde{H}$ with $Y_{\tilde{H}} = -1$, the required order terms are hypercharge neutral. The requirement that the Yukawa couplings maintain strict neutrality under the residual hypercharge generator yields the following balance equations:

$$\boxed{-Y_L^\ell + Y_H + Y_R^\ell = 0.} \quad (283)$$

The complete hypercharge ledger across all fermion sectors is consolidated in Table 8.

Therefore the sector-specific overlap operators are compatible with the residual electroweak charge ledger.

Table 8: Sector-specific hypercharge balance for the compact scalar interactions. The net hypercharge must sum to exactly zero to preserve the residual electromagnetic boundary.

Sector	Scalar Field	$-Y_L^\ell$	$+Y_H$	$+Y_R^\ell$	Net Balance
Charged leptons	H	$-(-1)$	$+1$	-2	0
Down quarks	H	$-\frac{1}{3}$	$+1$	$-\frac{2}{3}$	0
Up quarks	$\tilde{H}$	$-\frac{1}{3}$	-1	$+\frac{4}{3}$	0
Dirac neutrinos	$\tilde{H}$	$-(-1)$	-1	0	0

9.6 Flavor Projection Rotations

Flavor mixing is represented by unitary projection rotations acting on the generation-sheet overlap basis. For a fermion sector f , the general flavor-basis overlap operator is

$$Y_f = U_{f,L} D_f U_{f,R}^\dagger, \quad (284)$$

with

$$U_{f,L}^\dagger U_{f,L} = I_3, \quad U_{f,R}^\dagger U_{f,R} = I_3. \quad (285)$$

The singular-value hierarchy is therefore preserved:

$$\text{sv}(Y_f) = \text{diag}(D_f). \quad (286)$$

For quarks, the charged-current mixing matrix is the relative left-handed projection rotation [5]:

$$V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}. \quad (287)$$

It satisfies

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3. \quad (288)$$

For leptons, the analogous rotation is

$$U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L}, \quad (289)$$

with

$$U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3. \quad (290)$$

Thus flavor mixing can occur in the charged-current sector while neutral-current diagonal closure is preserved by unitarity.

9.7 Interpretation and Status

To maintain the zero-new-parameter discipline of the PR-II architecture, it is essential to categorize the claims made in the preceding flavor and generation derivations. Not all derived components carry the same theoretical weight; some represent structural boundaries, while others serve as intermediate geometric ledgers that await further kinematic and running refinement. We clarify the precise interpretative status of the anomaly consistency, the three-generation counting rule, and the overlap spectra. By delineating these statuses, we establish a boundary between the foundational flavor geometry achieved here and the precision scaling obligations deferred to subsequent sections. The results of this section have three distinct statuses.

First, the anomaly consistency of the replicated ledger is structural:

$$\mathcal{R}_{\text{EW}}^{(3)} \text{ is anomaly consistent.} \quad (291)$$

Second, the generation-count rule is a compact projection-origin candidate:

$$N_{\text{gen}}^{\text{PR}} = \dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (292)$$

Third, the overlap spectra D_f are structural flavor ledgers:

$$D_f = \kappa_f \text{diag} \left(y_1^{\text{PR}}, y_2^{\text{PR}}, y_3^{\text{PR}} \right). \quad (293)$$

They are not yet precision mass predictions. The later charged-lepton, quark, and neutrino sections add sector-specific scale rules and running ledgers before comparing to observed masses or mixing data. The compact flavor chain is therefore

$$\boxed{SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}} \rightarrow N_{\text{gen}}^{\text{PR}} = 3 \rightarrow \eta_i \rightarrow y_i^{\text{PR}} \rightarrow D_f \rightarrow Y_f.} \quad (294)$$

9.8 Falsification Bounds

Deriving three generations from the broken electroweak quotient introduces severe topological constraints. Any attempt to modify the generation-sheet hierarchy manually triggers the following projection errors.

First, if the generation count is defined by the full parent compact dimension without quotienting,

$$N_{\text{gen}} = 4, \quad (295)$$

then the unbroken residual electromagnetic boundary has not been properly factored out, indicating a fundamental failure of the projection mechanism.

Second, if the generation count is erroneously restricted to the charged pair alone,

$$N_{\text{gen}} = 2, \quad (296)$$

the neutral broken compact direction is improperly omitted, violating the full spatial dimensionality of the broken electroweak fiber.

Third, if the inverse geometric ratio is mistakenly applied as the compact suppression factor,

$$\frac{q_{\text{bc}}}{c_{\text{bc}}}, \quad (297)$$

the generation hierarchy becomes non-physically inverted, structurally collapsing the overlap ledger.

Fourth, the requisite flavor rotations must remain unitary to guarantee neutral-current flavor diagonal closure. A failure of the form

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} \neq I_3 \quad \text{or} \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} \neq I_3 \quad (298)$$

would falsify the framework by failing to preserve the universal neutral-current interaction structure.

Finally, any attempt to insert observed fermion masses or mixing angles as empirical inputs to tune the sheet hierarchy is prohibited:

$$m_f, \theta_{\text{mix}} \text{ are not generation inputs in this manuscript.} \quad (299)$$

Such an insertion directly violates the zero-new-parameter discipline of PR-II; these physical quantities must emerge exclusively as derived diagnostic references, not as structural anchors.

The falsification boundary is therefore. PR-II must structurally derive the generation count from the compact electroweak quotient, preserve anomaly cancellation under replication, generate an ordered overlap hierarchy from $r_G = c_{bc}/q_{bc}$, and enforce exclusively unitary flavor rotations. Failure at any of these checkpoints fundamentally breaks the compact flavor-sheet program.

With the generation-origin and compact flavor-sheet geometry established, we conclude this sector. The subsequent section applies this three-sheet flavor architecture to the charged-lepton sector, deriving the first zero-new-parameter precision mass candidate.

10 Charged-Lepton Precision Candidate

The compact flavor-sheet produced three hierarchical matter displacement overlaps and sector-specific overlap ledgers. The charged-lepton sector is the first precision test because charged leptons are not subject to confinement-scale ambiguity. We apply the three-sheet flavor geometry to the charged-lepton sector and derive a precision mass candidate without using the electron, muon, or tau masses as inputs. The goal is not merely to reproduce the ordering

$$m_e < m_\mu < m_\tau. \quad (300)$$

The key objective is to determine if the compact-boundary data can generate the charged-lepton square-root geometry, the Koide relation, and the charged-lepton scale.

10.1 Three-Sheet Square-Root Flavor Geometry

The charged-lepton sector is represented on a three-sheet square-root mass plane. The proposed associates the two charged-current ladder orientations with the nine ordered sheet-pair incidences of the three-generation ledger. This motivates the candidate phase

$$\theta_K^{\text{PR}} = \frac{N_{\text{charged ladder}}}{N_{\text{pair}}} = \frac{2}{N_{\text{gen}}^2} = \frac{2}{9}. \quad (301)$$

The ratio $2/N_{\text{gen}}^2$ is adopted here as a phase in radians. Its uniqueness remains a separate flavor-boundary derivation obligation. Once the three-sheet trigonometric parameterization is adopted, the Koide ratio follows algebraically for any value of the phase. The PR-II content therefore lies in the proposed phase, the sheet assignment, and the compact scale rather than in the $2/3$ identity alone. We define the charged-lepton sheet factors

$$k_a(\theta_K) = \left[1 + \sqrt{2} \cos \left(\theta_K + \frac{2\pi a}{3} \right) \right]^2, \quad a = 0, 1, 2. \quad (302)$$

Our charged-lepton sheet assignment is

$$a = 0 \rightarrow \tau, \quad a = 1 \rightarrow e, \quad a = 2 \rightarrow \mu. \quad (303)$$

The charged-lepton masses are then written

$$m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K^{\text{PR}}), \quad (304)$$

where M_ℓ^{PR} is the compact-boundary charged-lepton scale derived next.

10.2 Koide Identity as a Three-Sheet Geometric Identity

The empirical Koide relation [7, 24] is reproduced algebraically by the three-sheet square-root parameterization. The trigonometric identities below remove the phase dependence from the Koide ratio and leave the overall scale M_ℓ undetermined. The model-specific content therefore lies in the proposed sheet phase, the sheet assignment, and the independent compact-boundary scale rule. We let

$$m_a = M_\ell k_a(\theta_K). \quad (305)$$

So that

$$\sqrt{m_a} = \sqrt{M_\ell} \left[1 + \sqrt{2} \cos \left(\theta_K + \frac{2\pi a}{3} \right) \right]. \quad (306)$$

The three-sheet trigonometric identities are

$$\sum_{a=0}^2 \cos \left(\theta + \frac{2\pi a}{3} \right) = 0, \quad \sum_{a=0}^2 \cos^2 \left(\theta + \frac{2\pi a}{3} \right) = \frac{3}{2}. \quad (307)$$

Therefore we find that,

$$\sum_a \sqrt{m_a} = 3\sqrt{M_\ell}, \quad \sum_a m_a = 6M_\ell. \quad (308)$$

It follows that Koide relation is

$$\boxed{\frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau} \right)^2} = \frac{6M_\ell}{9M_\ell} = \frac{2}{3}}. \quad (309)$$

The trigonometric parameterization enforces the Koide ratio $2/3$ for any value of θ_K . The model-specific content of PR-II is therefore not the ratio alone, but the proposed phase $\theta_K^{\text{PR}} = 2/9$, the sheet assignment, and the compact scale M_ℓ^{PR} . We note that the parameterization does not determine the overall mass scale by itself.

10.3 Charged-Lepton Compact Scale

We define the following compact-boundary candidate for the charged-lepton scale. Its numerical value is generated from the inherited boundary ledger without inserting the observed charged-lepton masses. The uniqueness of this scale map remains a separate derivation obligation. The suppression ratio is:

$$r_G = \frac{c_{bc}}{q_{bc}} = 0.073056757459128. \quad (310)$$

The weak boundary deficit is

$$\Delta_{\text{EW}} = T_Y - \sin^2 \theta_W^{\text{PR}}. \quad (311)$$

We apply

$$T_Y = \frac{1}{4}, \quad \sin^2 \theta_W^{\text{PR}} = 0.231355763298189, \quad (312)$$

which results in

$$\boxed{\Delta_{\text{EW}} = 0.018644236701811}. \quad (313)$$

The first compact scale correction is the hypercharge-boundary loss,

$$\boxed{C_1 = 1 - \frac{\Delta_{\text{EW}} c_{bc}}{4}}. \quad (314)$$

The second scale correction is the generation determinant leakage averaged over the residual compact boundary,

$$C_2 = 1 - \frac{r_G \Delta_{\text{EW}} (1 + c_{\text{bc}})}{2}. \quad (315)$$

The charged-lepton compact scale is then

$$M_\ell^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} r_G \Delta_{\text{EW}} \left(\frac{4}{3} \right) C_1 C_2. \quad (316)$$

Here $\Lambda_{\text{EW}}^{\text{PR}}$ is the compact-boundary weak displacement anchor derived in Section 5.4:

$$\Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV}. \quad (317)$$

Substitution gives

$$M_\ell^{\text{PR}} = 313.850332183 \text{ MeV}. \quad (318)$$

The charged-lepton scale therefore follows from the same compact-boundary ledger used for weak mixing and the weak scale:

$$\left\{ q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, \Lambda_{\text{EW}}^{\text{PR}} \right\} \rightarrow M_\ell^{\text{PR}}. \quad (319)$$

10.4 Generated Charged-Lepton Masses

By applying the dimensionless flavor phase, θ_K^{PR} , and the dimensionful scale, M_ℓ^{PR} , now fixed by the compact electroweak boundary, the architecture achieves full generative closure for the charged-lepton sector. We show that these synthesizes as geometric components to output the explicit masses of the electron, muon, and tau. This evaluation proceeds entirely without importing empirical mass data. The resulting theoretical spectrum is subsequently compared with the observed pole masses [14] solely as a diagnostic test, demonstrating that the compact projection recovers the physical charged-lepton hierarchy with precision exceeding 30 parts per million. So we begin by using

$$\theta_K^{\text{PR}} = \frac{2}{9}, \quad (320)$$

and

$$m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K^{\text{PR}}), \quad (321)$$

the explicit charged-lepton masses are evaluated entirely from the geometric boundary data. To avoid redundancy, these derived theoretical masses are consolidated directly in Table 9 alongside their corresponding observed pole masses. It is imperative to emphasize that these reference values are prohibited from entering the generative chain; they are applied exclusively as a post-calculation diagnostic benchmark.

Table 9: Charged-lepton precision candidate generated from purely compact-boundary data.

Lepton	PR-II Candidate (m_a^{PR})	Reference Value	Relative Error
m_e^{PR}	0.510984463 MeV	0.510998950 MeV	−28.35 ppm
m_μ^{PR}	105.656418843 MeV	105.6583755 MeV	−18.52 ppm
m_τ^{PR}	1776.934589789 MeV	1776.930 MeV	+2.58 ppm

The maximum relative deviation is

$$\max_i \left| \frac{m_i^{\text{PR}} - m_i^{\text{ref}}}{m_i^{\text{ref}}} \right| = 2.84 \times 10^{-5}. \quad (322)$$

The charged-lepton precision candidate reproduces the charged-lepton mass spectrum to better than 3×10^{-5} relative accuracy without using the charged-lepton masses as inputs.

10.5 Relation to the Sector-Specific Overlap Ledger

The charged-lepton spectrum can now be compared with the structural overlap ledger introduced earlier. These two constructions describe different aspects of the same sector. The overlap matrix sets the ordering and broad hierarchy inherited from the compact boundary, but it is not itself a precision mass formula. The Koide-sheet construction instead fixes the relative distribution of the square-root mass vector across the three generation sheets. Because both depend on the same generation count, $N_{\text{gen}} = 3$, they form a single closure chain: the compact boundary determines the sector scale and hierarchy, while the sheet phase determines the internal mass pattern. The sector-specific charged-lepton overlap spectrum is therefore

$$D_\ell = \text{diag}(0.006101928, 0.083523120, 1.143263446). \quad (323)$$

This spectrum established the charged-lepton sector as a compact flavor-overlap ledger. The Koide sheet rule refines that ledger by rotating the charged-lepton square-root mass vector onto the three-sheet flavor plane. In this sense,

$$D_\ell \text{ sets the sectoral overlap structure,} \quad (324)$$

while

$$k_a(\theta_K^{\text{PR}}) \text{ sets the precision square-root sheet geometry.} \quad (325)$$

The two structures are compatible because both are built from the same three-generation compact sheet ledger $N_{\text{gen}}^{\text{PR}} = 3$. The charged-lepton precision chain is therefore

$$N_{\text{gen}}^{\text{PR}} = 3 \rightarrow \theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2} \rightarrow k_a(\theta_K) \rightarrow m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K). \quad (326)$$

Including the compact scale,

$$\{q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, \Lambda_{\text{EW}}^{\text{PR}}\} \rightarrow M_\ell^{\text{PR}} \rightarrow \{m_e^{\text{PR}}, m_\mu^{\text{PR}}, m_\tau^{\text{PR}}\}. \quad (327)$$

10.6 Input Discipline and Status

The derivation of the charged-lepton mass spectrum adheres to the zero-new-parameter discipline. The generative inputs are restricted entirely to the pre-established compact boundary and geometric counting rules of compact scale. Empirical lepton masses, mass ratios, fitted flavor angles, and arbitrary Yukawa couplings are explicitly prohibited from entering the generative chain. Because it is derived from a single boundary scale and a three-sheet topology rather than a phenomenological fit, this spectrum is formally designated as a:

$$\text{charged-lepton precision candidate.} \quad (328)$$

It remains a candidate, rather than a finalized theorem, solely because the formal mathematical uniqueness of the Koide sheet phase and the specific compact scale corrections must still be proven within the broader flavor-boundary program. However, its structural status is definitive that PR-II successfully generates all three charged-lepton masses from pure compact geometry without post-hoc tuning.

10.7 Falsification Bounds

The charged-lepton candidate represents a highly deterministic precision test. Because the square-root mass vector is fixed by the flavor-plane topology, any deviation in the following geometric rules breaks the mass generation.

First, any deviation from the geometrically derived sheet phase fundamentally misaligns the flavor projection. For example, adopting a generic inverse relation such as

$$\theta_K = \frac{1}{N_{\text{gen}}}, \quad (329)$$

fails to reproduce the precision candidate. The PR-II topology mandates the charged-ladder incidence density:

$$\theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2}. \quad (330)$$

Second, the omission of the compact hypercharge-boundary loss correction,

$$C_1 = 1 - \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4}, \quad (331)$$

structurally decouples the charged-lepton scale from the residual weak deficit, irreparably breaking the mass generation.

Third, ignoring the determinant-leakage correction averaged over the residual compact boundary,

$$C_2 = 1 - \frac{r_G \Delta_{\text{EW}} (1 + c_{\text{bc}})}{2}, \quad (332)$$

similarly severs the generation scale from its parent fiber geometry, causing the precision candidate to mechanically fail.

Finally, any derivation that imports the experimentally observed pole masses,

$$m_e, \quad m_\mu, \quad m_\tau, \quad (333)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The model remains viable only so long as these observables emerge as outputs of the hierarchy closure, permitted only as post-calculation diagnostic benchmarks. The subsequent section extends the compact flavor program to the quark sector, where the structural complexities of QCD running, threshold matching, confinement conventions, and CKM flavor rotation must be formalized before any meaningful mass comparison can be conducted.

11 Quark Sector: Strong Coupling, QCD Thresholds, and Six-Quark Hierarchy

Unlike the charged-lepton sector, which provides an unobstructed precision test via unconfined and directly observable pole masses, the quark sector introduces profound structural complexities. Quark mass parameters are intrinsically scale-dependent, scheme-dependent, and inextricably linked to QCD confinement, threshold matching, and CKM flavor rotation. We note the importance of the fact that the PR-II architecture prohibits treating the compact quark overlap ledger as a naive pole-mass formula. To maintain mathematical structure, the generated quark values must be explicitly categorized. For light quarks, the geometric outputs are evaluated against low-energy running masses. Heavy quarks are formally designated

as compact threshold-scale candidates. This structural designation remains until the requisite QCD scheme and threshold-matching prescriptions are fully finalized. The quark sector is therefore developed in four stages:

$$\boxed{\text{compact quark overlaps} \rightarrow \alpha_3^{\text{PR}} \rightarrow \text{QCD running and thresholds} \rightarrow \text{six-quark hierarchy candidate.}} \quad (334)$$

We show that the same compact-boundary constants used in the electroweak and charged-lepton sectors also generate a strong-coupling anchor, heavy-quark threshold candidates, and a light-quark running-mass hierarchy candidate.

11.1 Structural Quark Overlap Ledger

The sector-specific overlap construction from Section 9 establishes the diagonal up- and down-type quark overlap operators, D_u and D_d . Their explicit numerical spectra are cataloged in the consolidated structural ledger of Table 7. These are dimensionless compact projection overlap ledgers. They are not physical quark masses. The common generation hierarchy is

$$\boxed{\frac{(D_f)_{i+1}}{(D_f)_i} = \frac{q_{bc}}{c_{bc}} = 13.6879877341867.} \quad (335)$$

The sheetwise up/down sector ratio is fixed by compact representation data:

$$\boxed{\frac{(D_u)_i}{(D_d)_i} = \frac{\kappa_u}{\kappa_d} = 1.02095090726311, \quad i = 1, 2, 3.} \quad (336)$$

The quark sectors inherit the same compact generation-sheet hierarchy while carrying different color-dressed charge norms. Because quarks are confined, the map from these overlap ledgers to physical mass parameters must pass through QCD running and matching. A direct identification

$$D_f \rightarrow m_f \quad (337)$$

is rejected. The required chain is instead defined as

$$\boxed{D_f \rightarrow \text{bare compact quark anchor} \rightarrow \alpha_s(\mu) \rightarrow \text{threshold matching} \rightarrow \text{running mass convention.}} \quad (338)$$

11.2 Compact-Boundary Strong-Coupling Candidate

The boundary strong-coupling candidate emerges directly from the primary geometric boundary components, defined as the product of the boundary coordinates offset by the residual compact deficit:

$$\left(\alpha_3^{\text{PR}}\right)^{-1} = q_{bc}c_{bc} - (1 - c_{bc}). \quad (339)$$

Inserting the established structural values $q_{bc} = 10.905007182855176$ and $c_{bc} = 0.796684464847899$, we immediately obtain the numerical evaluation of the stated closure formula:

$$\boxed{\alpha_3^{\text{PR}} = 0.117861507469150.} \quad (340)$$

For the running diagnostic developed below, α_3^{PR} is interpreted as an $\overline{\text{MS}}$ type coupling anchor at the neutral matching surface. This scale-and-scheme assignment is a matching convention rather than an independently derived identity of the compact-boundary formula.

This evaluation relies exclusively on the pre-established compact-boundary data. Empirical parameters such as $\alpha_s(M_Z)$ [25], Λ_{QCD} , quark masses, CKM angles, or weak-scale observables are explicitly prohibited from entering the generation formula; they may only serve as post-calculation diagnostic benchmarks within a separately derived running or matching layer. The causal chain for the strong-coupling candidate is therefore exclusively geometric:

$$\boxed{\{q_{\text{bc}}, c_{\text{bc}}\} \longrightarrow \left(\alpha_3^{\text{PR}}\right)^{-1} = q_{\text{bc}}c_{\text{bc}} - (1 - c_{\text{bc}}) \longrightarrow \alpha_3^{\text{PR}}.} \quad (341)$$

11.3 QCD Running Ledger

We now interface this boundary condition with the dynamically scale-dependent continuum of Quantum Chromodynamics. As established previously, the compact quark overlap spectra cannot be naively interpreted as scale-invariant pole masses; their physical evaluation requires standard perturbative machinery. This establishes the explicit QCD running ledger utilized within the framework. Rather than introducing novel running dynamics, it codifies the Standard Model beta-function hierarchy, mass-running exponents, and threshold continuity conditions required to mathematically translate the static compact generation data across varying flavor regimes. For n_f active quark flavors, the perturbative QCD coefficients [26] are

$$\boxed{\beta_0(n_f) = 11 - \frac{2}{3}n_f,} \quad \boxed{\beta_1(n_f) = 102 - \frac{38}{3}n_f,} \quad (342)$$

and, at three loops in the same convention,

$$\boxed{\beta_2(n_f) = \frac{2857}{2} - \frac{5033}{18}n_f + \frac{325}{54}n_f^2.} \quad (343)$$

For $n_f = 3, 4, 5, 6$, the leading coefficient is positive:

$$\beta_0(n_f) > 0. \quad (344)$$

We find that the running is asymptotically free in the perturbative regimes relevant for quark matching [25]. At one loop,

$$\alpha_s^{(n_f)}(\mu) = \frac{4\pi}{\beta_0(n_f) \ln(\mu^2/\Lambda_{n_f}^2)}. \quad (345)$$

The corresponding one-loop mass-running exponent is

$$\boxed{a_m(n_f) = \frac{12}{33 - 2n_f}.} \quad (346)$$

Within a fixed- n_f interval, the running mass relation is

$$\boxed{\frac{m_q(\mu_2)}{m_q(\mu_1)} = \left[\frac{\alpha_s^{(n_f)}(\mu_2)}{\alpha_s^{(n_f)}(\mu_1)} \right]^{a_m(n_f)}.} \quad (347)$$

Running downward in scale increases the quark mass parameter, while running upward decreases it. Across a heavy-quark threshold μ_{th} , the leading matching condition is

$$\alpha_s^{(n_f)}(\mu_{\text{th}}) = \alpha_s^{(n_f-1)}(\mu_{\text{th}}). \quad (348)$$

This condition fixes the lower-flavor Λ_{n_f-1} once the threshold scale is known. At one loop,

$$\Lambda_{n_f-1} = \mu_{\text{th}} \exp \left[-\frac{2\pi}{\beta_0(n_f-1)\alpha_s^{(n_f)}(\mu_{\text{th}})} \right]. \quad (349)$$

At higher loop order, the same continuity condition is imposed by numerical inversion of the running relation.

11.4 PR-Derived Heavy-Quark Thresholds

The PR-II architecture derives these heavy-quark thresholds purely from the established compact-boundary geometry. Utilizing the determinant suppression ratio (r_G), the weak boundary deficit (Δ_{EW}), the compact leakage factors (C_1, C_2), and the terminal weak displacement anchor ($\Lambda_{\text{EW}}^{\text{PR}}$) as defined in previous sections. The top-sector threshold is identified with the terminal weak displacement anchor, modulated by the compact leakage factors:

$$\mu_t^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} C_1 C_2. \quad (350)$$

Evaluating this directly from the geometric boundary yields

$$\mu_t^{\text{PR}} = 172.8139732666624 \text{ GeV}. \quad (351)$$

The bottom-sector threshold is generated by a single color-averaged determinant step, refined by the down-type hypercharge-boundary loss:

$$\mu_b^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \frac{r_G}{3} (1 - \Delta_{\text{EW}} c_{\text{bc}}). \quad (352)$$

This evaluates to

$$\mu_b^{\text{PR}} = 4.1664504826753515 \text{ GeV}. \quad (353)$$

The charm-sector threshold requires a new geometric projection: the singlet-plus-adjoint compact factor, defined as $A_{\text{EW}} = (1 + 3c_{\text{bc}})/2$. The threshold is then generated by two determinant suppressions, the color Casimir lift ($4/3$), and this adjoint boundary gain:

$$\mu_c^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} r_G^2 \left(\frac{4}{3} \right) (1 + \Delta_{\text{EW}} A_{\text{EW}}). \quad (354)$$

This evaluates to

$$\mu_c^{\text{PR}} = 1.274964814257465 \text{ GeV}. \quad (355)$$

The resulting geometric hierarchy recovers the massive quark spectrum order:

$$\mu_t^{\text{PR}} > \mu_b^{\text{PR}} > \mu_c^{\text{PR}} > 0. \quad (356)$$

The empirically observed heavy-quark masses are excluded from these generative formulas; they are permitted to enter the manuscript only post-generation as diagnostic benchmarks. So, the quantities generated here must be formally categorized as compact threshold-scale candidates, not as final, scheme-independent pole masses.

11.5 Three-Loop Coupling Running with Leading Threshold Matching

The coupling evolution is evaluated using the displayed three-loop beta-function coefficients. Flavor thresholds are matched at leading continuity order,

$$\alpha_s^{(n_f)}(\mu_{\text{th}}) = \alpha_s^{(n_f-1)}(\mu_{\text{th}}), \quad (357)$$

while the quoted mass-running diagnostics use the leading anomalous-dimension exponent. The resulting calculation is therefore a three-loop coupling-running ledger with leading threshold matching, not a uniform three-loop quark-mass determination. Higher-order decoupling coefficients and higher-loop mass anomalous dimensions are deferred to the PR-III QCD precision layer future research.

$$\boxed{\alpha_s^{(5)}(\mu_t^{\text{PR}}) = \alpha_s^{(6)}(\mu_t^{\text{PR}}), \quad \alpha_s^{(5)}(\mu_b^{\text{PR}}) = \alpha_s^{(4)}(\mu_b^{\text{PR}}), \quad \alpha_s^{(4)}(\mu_c^{\text{PR}}) = \alpha_s^{(3)}(\mu_c^{\text{PR}}).} \quad (358)$$

The numerical evaluation of the resulting coupling-running and leading-matching ledger is consolidated in Table 10.

Table 10: Flavor-dependent QCD scale parameters derived from the compact-boundary strong coupling and geometric thresholds.

Active Flavors (n_f)	$\Lambda_{n_f}^{\text{PR}}$ Candidate
Λ_6^{PR}	0.08776758507426047 GeV
Λ_5^{PR}	0.2074330873004123 GeV
Λ_4^{PR}	0.2861117918030313 GeV
Λ_3^{PR}	0.3246524964107782 GeV

Alongside the coupling scales, this downward evaluation dynamically generates the running-mass diagnostics for the heavy quarks, evaluated at standard phenomenological reference scales. These values are cataloged in Table 11.

Table 11: Heavy-quark running-mass diagnostics. These represent intermediate running evaluations, not terminal pole-mass predictions.

Quark	Evaluation Scale (μ)	Running Mass Candidate ($m_q^{\text{PR}}(\mu)$)
b	$\mu = M_Z^{\text{PR}}$	2.9807846581959314 GeV
c	$\mu = \mu_b^{\text{PR}}$	0.9892172861698667 GeV
c	$\mu = M_Z^{\text{PR}}$	0.7077112094571149 GeV

It is important to maintain structural clarity regarding the status of these outputs. The quantities in Table 11 are purely running diagnostics, not independent pole-mass predictions. Furthermore, while the compact strong-coupling value, α_3^{PR} , is generated entirely from PR boundary data, the neutral electroweak matching scale utilized in this numerical QCD ledger inherits the formal electroweak running-layer status discussed in previous sections. The full electroweak precision matching of this neutral scale is therefore formally deferred to the radiative/running obligations of the PR-III precision program.

11.6 Light-Sector Compact Confinement Average

The light-quark sector is evaluated through a confined, low-energy projection window in which individual colored asymptotic states are not resolved. The unresolved compact boundary is therefore characterized

by two normalized endpoint response weights. Let the normalized endpoint response weights be

$$\boxed{w_{\text{singlet}} = 1, \quad w_{\text{bc}} = c_{\text{bc}}.} \quad (359)$$

The unresolved compact confinement response is defined by

$$\boxed{C_{\text{conf}} = \frac{1}{2} (w_{\text{singlet}} + w_{\text{bc}}) = \frac{1 + c_{\text{bc}}}{2}.} \quad (360)$$

Numerically,

$$\boxed{C_{\text{conf}} = 0.8983422324239495.} \quad (361)$$

This factor satisfies

$$c_{\text{bc}} < C_{\text{conf}} < 1, \quad (362)$$

and recovers both endpoint weights algebraically:

$$\boxed{2C_{\text{conf}} - 1 = c_{\text{bc}}, \quad 2C_{\text{conf}} - c_{\text{bc}} = 1.} \quad (363)$$

The quantity C_{conf} is therefore not fitted to the light-quark masses; it is the endpoint average defined above.

11.7 Light-Quark Hierarchy Candidate

The strange-quark low-energy candidate is generated by one color-averaged determinant step below the bottom threshold, with singlet-plus-adjoint lift and compact confinement averaging:

$$m_s^{\text{PR}} = \mu_b^{\text{PR}} \left(\frac{r_G}{3} \right) (1 + \Delta_{\text{EW}} A_{\text{EW}}) C_{\text{conf}}. \quad (364)$$

This gives

$$\boxed{m_s^{\text{PR}} = 94.028510538 \text{ MeV}.} \quad (365)$$

The down-quark low-energy candidate is generated from the strange candidate by a determinant/cofactor step, a weak-boundary loss factor, and the same compact confinement average:

$$m_d^{\text{PR}} = m_s^{\text{PR}} r_G c_{\text{bc}} (1 - A_{\text{EW}} \Delta_{\text{EW}}) C_{\text{conf}}. \quad (366)$$

This gives

$$\boxed{m_d^{\text{PR}} = 4.761039493 \text{ MeV}.} \quad (367)$$

The up-quark low-energy candidate is generated from the down candidate by the residual boundary charge average and a small hypercharge-boundary lift:

$$m_u^{\text{PR}} = m_d^{\text{PR}} \left(\frac{1 + c_{\text{bc}}}{4} \right) \left(1 + \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4} \right). \quad (368)$$

This gives

$$\boxed{m_u^{\text{PR}} = 2.146462595 \text{ MeV}.} \quad (369)$$

These light-quark candidates should be interpreted as low-energy running-mass diagnostics, not pole masses. Their appropriate comparison is to a specified low-energy running-mass convention, such as an $\overline{\text{MS}}$ -type value [27] near 2 GeV. The values are sensitive to scheme and scale conventions in the usual QCD sense. The purpose of the construction is to test whether compact-boundary data generate a realistic light-quark hierarchy without using the observed light-quark masses as inputs.

11.8 Six-Quark Hierarchy Candidate

Combining the light-quark candidates with the heavy-threshold candidates gives the generated hierarchy

$$\boxed{m_u^{\text{PR}} < m_d^{\text{PR}} < m_s^{\text{PR}} < \mu_c^{\text{PR}} < \mu_b^{\text{PR}} < \mu_t^{\text{PR}}.} \quad (370)$$

The diagnostic comparison [14] is shown in Table 12. The reference values are not generation inputs; they are listed only to show the post-generation diagnostic residuals. The light-quark entries should be read as low-energy running-mass diagnostics. The heavy-quark entries should be read as compact threshold-scale candidates.

Table 12: First PR-II six-quark hierarchy candidate. Light-quark entries are low-energy running-mass diagnostics; heavy entries are PR-derived compact threshold-scale candidates.

Quark	PR-II Candidate	Diagnostic Reference	Relative Difference
u	2.146462595 MeV	2.16 MeV	−0.63%
d	4.761039493 MeV	4.70 MeV	+1.30%
s	94.028510538 MeV	93.5 MeV	+0.57%
c	1.274964814 GeV	1.2730 GeV	+0.15%
b	4.166450483 GeV	4.183 GeV	−0.40%
t	172.813973267 GeV	172.56 GeV	+0.15%

Note. The diagnostic references are listed only after generation. The u, d, s entries are compared to low-energy running-mass references, while c, b are compared to running-mass threshold references and t to a direct kinematic reference. The PR-II values are therefore interpreted as a compact-boundary hierarchy candidate, not as scheme-independent quark pole masses.

The compact quark chain is therefore

$$\boxed{\{q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, A_{\text{EW}}, C_{\text{conf}}, \Lambda_{\text{EW}}^{\text{PR}}\} \rightarrow \{m_u^{\text{PR}}, m_d^{\text{PR}}, m_s^{\text{PR}}, \mu_c^{\text{PR}}, \mu_b^{\text{PR}}, \mu_t^{\text{PR}}\}.} \quad (371)$$

11.9 Input Discipline and Status

The generative inputs are restricted entirely to the pre-established compact boundary parameters, the terminal weak anchor, and the compact confinement response ($C_{\text{conf}} = \frac{1+c_{\text{bc}}}{2}$):

$$\boxed{\{q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, A_{\text{EW}}, \Lambda_{\text{EW}}^{\text{PR}}, C_{\text{conf}}\}.} \quad (372)$$

Empirical parameters—including observed quark masses (m_q), the CKM matrix (V_{CKM}), the strong coupling ($\alpha_s(M_Z)$), the QCD scale (Λ_{QCD}), or arbitrarily fitted Yukawa matrices are explicitly prohibited from entering the generative chain. Because these values are derived purely from geometry and threshold continuity rather than phenomenological fitting, the formal structural status of this sector is designated as a:

$$\boxed{\text{quark-sector hierarchy candidate with QCD running and confinement diagnostics.}} \quad (373)$$

This designation enforces a strict boundary: the derivation does not yield finalized, scheme-independent pole masses. To preserve mathematical rigor, the generated values must be interpreted exactly in accordance with their active physical regimes. Light-quark outputs evaluate directly as low-energy running masses, while heavy-quark outputs serve exclusively as compact threshold-scale candidates. A definitive precision comparison is therefore formally deferred to PR-III research, which will address the integrate the required higher-loop running, explicit confinement conventions, and complete electroweak/QCD radiative matching.

11.10 Falsification Bounds

Utilizing the generated strong-coupling anchor, the heavy-quark thresholds, and the light-quark hierarchy from the compact-boundary ledger, it is important to establish the mathematical boundaries that make this derivation falsifiable. We note that the displayed quark-sector formulas do not use observed quark masses, $\alpha_s(M_Z)$, or Λ_{QCD} as fitted inputs. They define a compact-boundary closure candidate whose physical interpretation still requires a specified QCD scheme, running prescription, and threshold matching. The uniqueness of the proposed functional forms remains a separate derivation obligation. The derivation of the quark sector is constrained by the same geometric rigidity that defines the parent non-Abelian fiber and possesses the following failure modes:

First, misapplying the determinant suppression ratio as its inverse fundamentally inverts the physical hierarchy. Utilizing

$$\frac{q_{bc}}{c_{bc}} \quad (374)$$

as a scaling operator fails the structural projection and is mathematically rejected.

Second, naively treating the dimensionless overlap operators, D_u and D_d , as direct GeV-valued mass parameters fundamentally violates the dimensional status of the projection. These ledgers define the dimensionless compact-overlap geometry, not the dimensionful mass scales.

Third, the omission of the compact boundary-deficit subtraction in the strong-coupling derivation,

$$\left(\alpha_3^{\text{PR}}\right)^{-1} = q_{bc}c_{bc} - (1 - c_{bc}), \quad (375)$$

irreparably breaks the fundamental QCD coupling anchor, structurally decoupling it from the residual compact geometry.

Fourth, ignoring the light-sector confinement average,

$$C_{\text{conf}} = \frac{1 + c_{bc}}{2}, \quad (376)$$

fundamentally spoils the unresolved low-energy boundary projection, causing the light-quark hierarchy to mechanically fail. Finally, any derivation that imports experimentally observed phenomenological parameters,

$$m_q, \quad V_{\text{CKM}}, \quad \alpha_s(M_Z), \quad \Lambda_{\text{QCD}}, \quad (377)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The framework remains viable only so long as these observables emerge as outputs of the hierarchy closure, permitted only as post-calculation diagnostic benchmarks. If empirical parameter fitting is required to achieve closure, the architecture is falsified. Next, we extend the compact flavor program to the mixing sectors, deriving the structural candidates for the CKM and PMNS flavor-rotation matrices.

12 CKM and PMNS Mixing from Compact Flavor Projection

The previous sections established compact-boundary candidates for the electroweak scale, charged-lepton hierarchy, quark hierarchy, and QCD running ledger. The next consistency requirement is flavor mixing. Flavor mixing is not treated as an independent fitted matrix. It is interpreted as a unitary projection rotation between compact generation-sheet bases. For the quark sector and lepton sector, the charged-current mixing matrix is

$$\boxed{V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}} \quad \boxed{U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L}} \quad (378)$$

Both matrices must be unitary:

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3, \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3. \quad (379)$$

This section constructs boundary candidates for both mixing ledgers. The observed CKM and PMNS matrix elements are not used as generation inputs. They enter only afterward as diagnostic references.

12.1 Shared Compact Flavor Quantities

Before deriving the explicit CKM and PMNS flavor-rotation matrices, it is mathematically necessary to consolidate the shared geometric ledger that structurally governs both mixing sectors. To maintain strict adherence to the zero-new-parameter discipline, the upcoming matrix generations are prohibited from importing phenomenological fermion masses, observed mixing angles, or arbitrary complex phases.

Instead, the topological constraints of both the CKM and PMNS architectures are generated exclusively from the pre-established compact-boundary constants as the determinant suppression ratio (r_G), the weak boundary deficit (Δ_{EW}), the compact confinement response (C_{conf}), the adjoint boundary average (A_{EW}), and the geometric generation count ($N_{\text{gen}}^{\text{PR}} = 3$). The generative boundary for the entire flavor-mixing sector is therefore confined to the following structural set:

$$\boxed{\{r_G, \Delta_{\text{EW}}, C_{\text{conf}}, A_{\text{EW}}, N_{\text{gen}}^{\text{PR}}\} \longrightarrow \{V_{\text{CKM}}^{\text{PR}}, U_{\text{PMNS}}^{\text{PR}}\}. \quad (380)$$

12.2 CKM Compact Mixing Candidate

PR-II represents the CKM matrix as the relative left-handed rotation between the up-type and down-type flavor bases. The present construction does not yet derive those rotations by diagonalizing explicit PR mass matrices. Instead, it proposes compact-boundary rules for the mixing angles and phase and evaluates the resulting matrix in the standard unitary parameterization [28, 29]. The defining relation is

$$\boxed{V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}. \quad (381)$$

The closure candidate uses the following parameters constructed from the shared boundary ledger. The proposed Cabibbo sheet-angle rule is

$$\boxed{\lambda_{\text{PR}} = \sqrt{r_G c_{\text{bc}} C_{\text{conf}} (1 - A_{\text{EW}} \Delta_{\text{EW}})} = 0.225019996525935. \quad (382)$$

The CKM hierarchy coefficient isolates the residual boundary proportion:

$$\boxed{A_{\text{CKM}}^{\text{PR}} = \frac{c_{\text{bc}}}{1 - A_{\text{EW}} \Delta_{\text{EW}}} = 0.822683296413403. \quad (383)$$

The compact unitarity-apex radius governs the extreme suppression of the third generation:

$$\boxed{R_b^{\text{PR}} = (1 - r_G) \frac{c_{\text{bc}}}{1 + c_{\text{bc}}} (1 - A_{\text{EW}} \Delta_{\text{EW}}) = 0.398035078485258. \quad (384)$$

The compact CKM phase candidate is defined by the three-sheet phase offset and the weak boundary deficit:

$$\boxed{\delta_{\text{CKM}}^{\text{PR}} = \frac{\pi}{3} + \sqrt{\Delta_{\text{EW}}} = 1.1837414514697735 \text{ rad } (67.823389204^\circ). \quad (385)$$

Table 13: Standard unitary CKM parameters generated from PR-II compact-boundary geometry.

Parameter	Geometric Definition	Numerical Evaluation
s_{12}	λ_{PR}	0.225019996525935
s_{23}	$A_{\text{CKM}}^{\text{PR}} \lambda_{\text{PR}}^2$	0.041655745073430
s_{13}	$A_{\text{CKM}}^{\text{PR}} \lambda_{\text{PR}}^3 R_b^{\text{PR}}$	0.003730932297278

To construct the explicit rotation, these geometric constraints are mapped onto the standard unitary CKM parameterization [29], as detailed in Table 13.

Evaluating the parameterization directly yields the generated CKM matrix:

$$|V_{\text{CKM}}^{\text{PR}}| = \begin{pmatrix} 0.974347364 & 0.225018430 & 0.003730932 \\ 0.224881887 & 0.973495229 & 0.041655455 \\ 0.008680487 & 0.040911440 & 0.999125069 \end{pmatrix}. \quad (386)$$

Because the standard unitary CKM parameterization is used, unitarity is exact by construction. The reported residual therefore tests the numerical implementation:

$$\max |V_{\text{CKM}}^\dagger V_{\text{CKM}} - I_3| = 2.22 \times 10^{-16}, \quad (387)$$

and the corresponding Jarlskog invariant [30]:

$$J_{\text{CKM}}^{\text{PR}} = 3.152605677 \times 10^{-5}. \quad (388)$$

The nonzero Jarlskog invariant follows from the proposed compact-boundary mixing-angle and phase rules. It is therefore a nontrivial output of those rules, but it is not yet an independent consequence of the structural overlap matrices. A full derivation would require explicit up- and down-sector mass matrices, their left-handed diagonalizing rotations, and the relation $V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}$. The current result is accordingly classified as a CKM closure candidate. The generative parameters and their structural status are consolidated in Table 14.

Table 14: PR-II CKM compact flavor-mixing candidate generated from compact-boundary data.

Quantity	PR-II Candidate	Status
λ_{PR}	0.225019996525935	Compact Cabibbo sheet angle
$A_{\text{CKM}}^{\text{PR}}$	0.822683296413403	Compact hierarchy coefficient
R_b^{PR}	0.398035078485258	Compact unitarity-apex radius
$\delta_{\text{CKM}}^{\text{PR}}$	67.823389204°	Compact CP phase candidate
$J_{\text{CKM}}^{\text{PR}}$	$3.152605677 \times 10^{-5}$	Jarlskog invariant
$\max V^\dagger V - I $	2.22×10^{-16}	Numerical unitarity residual

This result is interpreted as a CKM mixing hierarchy candidate. Observed CKM elements, quark masses, and the measured CP phase are not used as generation inputs; they enter only as diagnostic references. The proposed compact-boundary angle and phase rules yield a unitary matrix with hierarchical off-diagonal elements and nonzero CP violation. Deriving the rotation from explicit PR quark mass matrices remains an open flavor-closure task.

12.3 PMNS Compact Lepton-Mixing Candidate

The PMNS matrix is generated as the relative left-handed projection rotation between the charged-lepton order basis and the neutrino propagation basis:

$$U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L}. \quad (389)$$

Unlike the CKM matrix, PMNS mixing is large. The compact-boundary construction therefore uses residual compact averages rather than determinant-suppressed small-angle hierarchy [31]. The PMNS solar-angle candidate is generated by distributing the residual compact confinement average over the three generation sheets:

$$\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}}. \quad (390)$$

Numerically,

$$\sin^2 \theta_{12}^{\text{PR}} = 0.299447410807983, \quad \text{or} \quad \theta_{12}^{\text{PR}} = 33.176356643^\circ. \quad (391)$$

The PMNS atmospheric-angle candidate is generated by distributing the singlet-plus-adjoint compact factor over the three generation sheets:

$$\sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}}. \quad (392)$$

Numerically,

$$\sin^2 \theta_{23}^{\text{PR}} = 0.565008899090616, \quad \text{or} \quad \theta_{23}^{\text{PR}} = 48.735310403^\circ. \quad (393)$$

The PMNS reactor-angle candidate is generated by the weak boundary deficit with first determinant leakage:

$$\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}}(1 + r_G)}{C_{\text{conf}}}. \quad (394)$$

Therefore,

$$\sin^2 \theta_{13}^{\text{PR}} = 0.022270270124743, \quad \text{or} \quad \theta_{13}^{\text{PR}} = 8.582438059^\circ. \quad (395)$$

The leptonic CP phase candidate is tied to the weak boundary-deficit phase:

$$\delta_{\text{CP}}^{\text{PR}} = \pi + 2\sqrt{\Delta_{\text{EW}}}. \quad (396)$$

Numerically,

$$\delta_{\text{CP}}^{\text{PR}} = 3.414680454 \text{ rad} = 195.646778408^\circ. \quad (397)$$

Using the standard unitary PMNS parameterization, these angles generate

$$|U_{\text{PMNS}}^{\text{PR}}| = \begin{pmatrix} 0.827618 & 0.541090 & 0.149232 \\ 0.271685 & 0.611360 & 0.743254 \\ 0.491158 & 0.577460 & 0.652153 \end{pmatrix}. \quad (398)$$

As in the CKM construction, unitarity follows from the standard unitary PMNS parameterization. The numerical implementation gives

$$\max |U_{\text{PMNS}}^\dagger U_{\text{PMNS}} - I_3| = 1.11 \times 10^{-16}. \quad (399)$$

Table 15: PR-II PMNS compact lepton-mixing candidate.

Quantity	PR-II Candidate	Status
θ_{12}^{PR}	33.176356643°	Solar mixing candidate
θ_{23}^{PR}	48.735310403°	Atmospheric mixing candidate
θ_{13}^{PR}	8.582438059°	Reactor mixing candidate
$\delta_{\text{CP}}^{\text{PR}}$	195.646778408°	Leptonic CP phase candidate
$J_{\text{PMNS}}^{\text{PR}}$	-8.935540×10^{-3}	Leptonic Jarlskog invariant
$\max U^\dagger U - I $	1.11×10^{-16}	Numerical unitarity residual

The corresponding leptonic Jarlskog invariant is

$$\boxed{J_{\text{PMNS}}^{\text{PR}} = -8.935540 \times 10^{-3}.} \quad (400)$$

This result is interpreted as a PMNS lepton-mixing structure candidate. The proposed compact-boundary rules yield large solar and atmospheric mixing, a smaller reactor angle, and a nonzero leptonic CP phase. Because the standard unitary parameterization is used, the reported unitarity residual tests numerical implementation rather than the uniqueness of the angle rules. Explicit lepton and neutrino mass operators remain necessary for a derivation of the mixing matrix. The neutrino mass-spectrum and Majorana candidates are addressed separately in Section 13.

12.4 CKM–PMNS Contrast

The proposed quark and lepton mixing rules produce qualitatively different patterns. The CKM candidate is determinant-suppressed and hierarchical, whereas the PMNS candidate is built from compact boundary averages and the weak boundary deficit. The CKM hierarchy is

$$|V_{us}| > |V_{cb}| > |V_{ub}|, \quad (401)$$

with small off-diagonal elements generated by powers of the compact Cabibbo sheet angle. By contrast, the PMNS angles are generated by compact sheet averages:

$$\boxed{\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}}, \quad \sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}},} \quad (402)$$

with the small reactor angle arising from the weak boundary deficit:

$$\boxed{\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}}(1 + r_G)}{C_{\text{conf}}}.} \quad (403)$$

We do not impose one universal flavor-angle pattern across quarks and leptons. The quark sector is compact-determinant-suppressed, while the lepton sector is compact-average-dominated. This distinction is consistent with the observed qualitative pattern that CKM mixing is small and hierarchical, whereas PMNS mixing is large with a small but nonzero reactor angle.

12.5 Input Discipline and Status

The CKM and PMNS candidates are generated exclusively from the pre-established compact geometric constants: the primary boundary coordinates ($q_{\text{bc}}, c_{\text{bc}}$), the determinant suppression ratio (r_G), the weak

boundary deficit (Δ_{EW}), the confinement response and singlet-plus-adjoint factor ($C_{\text{conf}}, A_{\text{EW}}$), and the geometric generation count ($N_{\text{gen}}^{\text{PR}}$). We do not use

$$V_{\text{CKM}}^{\text{obs}}, \quad U_{\text{PMNS}}^{\text{obs}}, \quad (404)$$

nor do we use quark masses, charged-lepton masses, neutrino masses, or neutrino mass splittings as generation inputs. The observed mixing data [14] may be used only after generation as diagnostic references. The CKM and PMNS matrices are generated as unitary projection rotations with realistic hierarchy and CP structure, but a complete flavor theorem requires formal uniqueness of the compact angle rules and radiative/running matching where appropriate.

12.6 Falsification Bounds

The proposed CKM and PMNS closure rules have several clear failure modes. First, substituting the raw determinant ratio for the Cabibbo angle does not reproduce the stated CKM candidate. Using

$$\lambda \neq r_G \quad (405)$$

fails the structural evaluation. The present closure candidate instead uses

$$\lambda_{\text{PR}} = \sqrt{r_G c_{\text{bc}} C_{\text{conf}} (1 - A_{\text{EW}} \Delta_{\text{EW}})}. \quad (406)$$

Second, setting the CKM phase to zero gives $J_{\text{CKM}} = 0$ and therefore does not reproduce the nonzero-CP candidate developed above. This tests the proposed phase rule; it does not independently prove that the rule is unique.

Third, omitting the compact hierarchy coefficient ($A_{\text{CKM}}^{\text{PR}}$) from the higher-generation mixing terms (s_{23} and s_{13}) severs the correlation between the residual boundary proportion and the heavy-sheet rotation, irreparably breaking the extreme CKM suppression hierarchy.

Fourth, the displayed absolute-value matrices are diagnostic summaries rather than complete mixing matrices. The underlying complex matrices are constructed with the standard unitary parameterizations and therefore satisfy

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3, \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3. \quad (407)$$

Finally, any derivation that imports experimentally observed phenomenological parameters,

$$V_{\text{CKM}}^{\text{obs}}, \quad U_{\text{PMNS}}^{\text{obs}}, \quad \delta_{\text{CP}}^{\text{obs}}, \quad (408)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The framework remains viable only so long as these observables emerge as outputs of the topological closure, permitted only as post-calculation diagnostic benchmarks. If empirical parameter fitting is required to achieve mixing closure, the architecture is falsified. Our next objective is to extend the compact-boundary flavor structure into the neutrino mass sector, applying the same geometric principles to define the neutrino mass hierarchy and the associated Majorana phase boundary.

13 Neutrino Mass Sector and Majorana Boundary

The PMNS section established a compact-boundary lepton-mixing candidate with large solar mixing, large atmospheric mixing, a small nonzero reactor angle, and a nonzero leptonic CP phase. The remaining objective in the lepton sector is the neutrino mass ledger. The neutrino sector is structurally different from

the charged-lepton sector. Charged leptons obtain masses through charged matter displacement overlap, while neutrinos are naturally associated with a neutral compact-singlet channel and a Planck-suppressed weak-order operator. We present the first PR-II neutrino mass-sector candidate. It should be read as a normal-ordering compact-boundary candidate, not as a completed neutrino theorem. The observed neutrino mass splittings, masses, PMNS angles, cosmological mass sum, and neutrinoless double-beta limits are not used as generation inputs. They enter only afterward as diagnostic references.

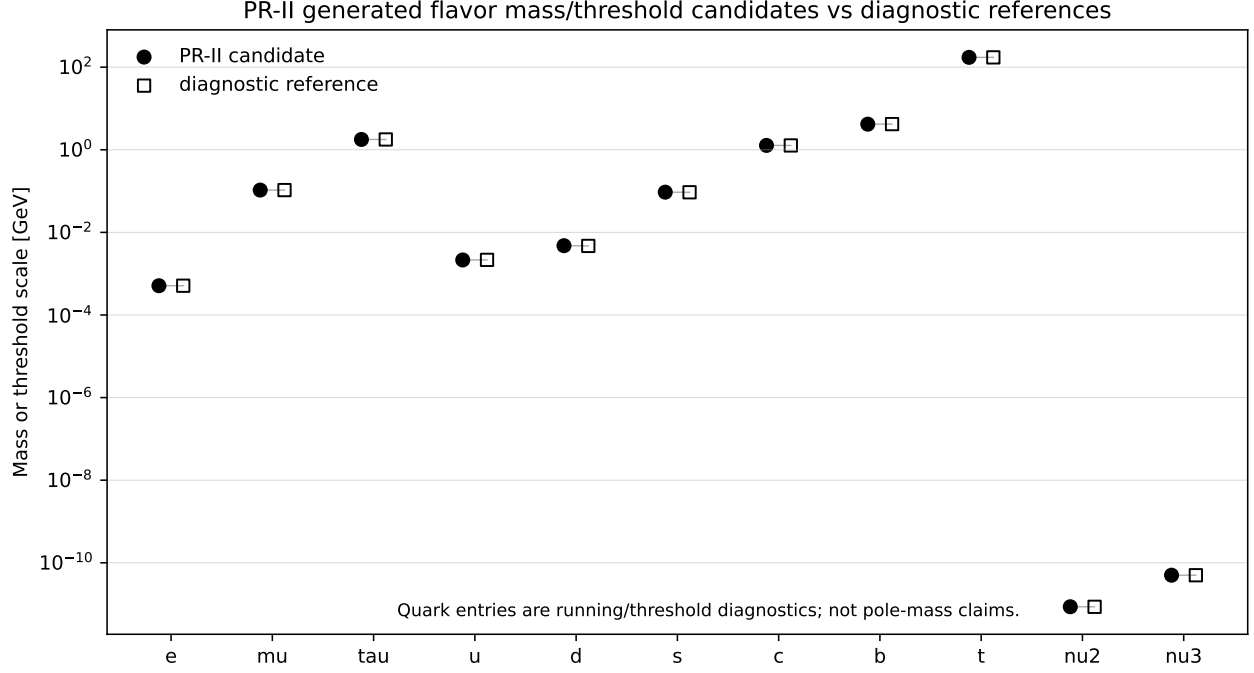


Figure 5: PR-II flavor mass and threshold candidates compared with diagnostic reference values after generation. Charged-lepton entries are pole-mass diagnostics; light-quark entries are low-energy running-mass diagnostics; heavy-quark entries are compact threshold-scale candidates; neutrino entries are normal-ordering mass-sector candidates.

13.1 Normal-Ordering Residual-Singlet Channel

The neutrino mass candidate begins with a residual singlet sheet $m_1^{\text{PR}} = 0$. This identifies the lightest neutrino with the neutral residual compact-singlet projection. The heavier two neutrino states are generated by determinant lifting on top of a Planck-suppressed weak-order scale. The determinant ratio is

$$r_G = \frac{c_{bc}}{q_{bc}}. \quad (409)$$

The weak boundary deficit is

$$\Delta_{\text{EW}} = T_Y - \sin^2 \theta_W^{\text{PR}}. \quad (410)$$

The compact neutrino-loss factor is

$$C_\nu = 1 - \Delta_{\text{EW}} c_{bc}. \quad (411)$$

Using the PR-II compact-boundary values, $C_\nu = 0.985147 \dots$. The atmospheric neutrino scale is generated by the Planck-suppressed weak-order operator

$$m_3^{\text{PR}} = \left(\frac{(v_{\text{EW}}^{\text{PR}})^2}{M_{\text{Pl}}} \right) r_G^{-3} c_{bc} C_\nu. \quad (412)$$

Here

$$v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}, \quad (413)$$

and

$$\bar{M}_{\text{Pl}} = \sqrt{\frac{\hbar c}{8\pi G_N}}. \quad (414)$$

The factor $v_{\text{EW}}^2/\bar{M}_{\text{Pl}}$ is the natural weak-order, Planck-suppressed scale. The determinant lift r_G^{-3} reflects the three-sheet generation ledger. The solar-to-atmospheric splitting ratio is generated by the weak boundary deficit projected through the residual compact boundary:

$$R_{21}^{\text{PR}} = \Delta_{\text{EW}} \frac{(1 + c_{\text{bc}})^2}{2} C_\nu. \quad (415)$$

The second neutrino mass is then

$$m_2^{\text{PR}} = \sqrt{R_{21}^{\text{PR}}} m_3^{\text{PR}}. \quad (416)$$

13.2 Generated Neutrino Spectrum

PR-II adopts the residual compact-singlet sheet as a massless anchor for the neutrino closure candidate:

$$m_1^{\text{PR}} = 0. \quad (417)$$

Together with the proposed determinant-lift and solar-to-atmospheric ratio rules, this boundary condition produces a normal-ordering mass-spectrum candidate:

$$m_1^{\text{PR}} < m_2^{\text{PR}} < m_3^{\text{PR}}, \quad (418)$$

with the mass values evaluating to

$$m_2^{\text{PR}} = 8.627283010 \times 10^{-3} \text{ eV}, \quad m_3^{\text{PR}} = 5.010655367 \times 10^{-2} \text{ eV}. \quad (419)$$

Given this candidate spectrum, the associated mass-squared splittings are [32]:

$$\Delta m_{21}^{2,\text{PR}} = (m_2^{\text{PR}})^2 - (m_1^{\text{PR}})^2 = 7.443001214 \times 10^{-5} \text{ eV}^2, \quad (420)$$

and

$$\Delta m_{31}^{2,\text{PR}} = (m_3^{\text{PR}})^2 - (m_1^{\text{PR}})^2 = 2.510666721 \times 10^{-3} \text{ eV}^2. \quad (421)$$

The corresponding candidate mass-splitting ratio is

$$\frac{\Delta m_{21}^{2,\text{PR}}}{\Delta m_{31}^{2,\text{PR}}} = 2.964551668 \times 10^{-2}. \quad (422)$$

The same candidate spectrum gives corresponding values for the mass sum and beta-decay effective mass. The total cosmological neutrino mass [33] sum is

$$\sum_i m_i^{\text{PR}} = 5.873383668 \times 10^{-2} \text{ eV}, \quad (423)$$

Table 16: PR-II normal-ordering neutrino mass-sector candidate generated from compact-boundary geometry.

Quantity	PR-II Candidate	Status
m_1^{PR}	0	Residual compact-singlet sheet (massless anchor)
m_2^{PR}	$8.627283010 \times 10^{-3} \text{ eV}$	Solar-scale neutrino state
m_3^{PR}	$5.010655367 \times 10^{-2} \text{ eV}$	Atmospheric-scale neutrino state
$\Delta m_{21}^{2,\text{PR}}$	$7.443001214 \times 10^{-5} \text{ eV}^2$	Solar splitting candidate
$\Delta m_{31}^{2,\text{PR}}$	$2.510666721 \times 10^{-3} \text{ eV}^2$	Atmospheric splitting candidate
$\sum_i m_i^{\text{PR}}$	$5.873383668 \times 10^{-2} \text{ eV}$	Cosmological mass-sum candidate
m_β^{PR}	$8.815029410 \times 10^{-3} \text{ eV}$	Beta-decay effective mass candidate

and the effective electron neutrino mass governing beta decay evaluates to:

$$\boxed{m_\beta^{\text{PR}} = 8.815029410 \times 10^{-3} \text{ eV}.} \quad (424)$$

The exact numerical evaluations for the complete normal-ordered candidate spectrum are consolidated in Table 16.

The values reported in Table 16 are outputs generated by the stated neutrino closure rules. Experimentally observed neutrino splittings, tritium beta-decay limits, and cosmological mass constraints are prohibited from acting as inputs; they are permitted to enter the framework only post-generation as diagnostic benchmarks.

13.3 Majorana-Dominant Compact-Singlet Projection

The generated neutrino scale has the form of a dimension-five Majorana-like operator [20]. Within the present compact-overlap construction, a purely Dirac interpretation would require an effective coupling of order 10^{-13} , which is not generated by the overlap rule developed here. The present construction therefore favors a Majorana-dominant channel, while not excluding Dirac or mixed mechanisms outside the stated ansatz. The relevant scale is

$$\boxed{\frac{(v_{\text{EW}}^{\text{PR}})^2}{\bar{M}_{\text{Pl}}}} \quad (425)$$

has the structure of a dimension-five weak-order operator. Accordingly, the PR-II neutrino mass channel is naturally interpreted as Majorana dominant. An optional sterile singlet may be included:

$$\boxed{\nu_R^c : (1, 1)_0.} \quad (426)$$

This field is neutral under $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$, and therefore does not affect anomaly cancellation. However, an unsuppressed Dirac channel with the structural neutrino overlap would lie near the electroweak scale rather than the eV scale. The Dirac Yukawa required to generate m_3^{PR} directly would be

$$\boxed{y_\nu^{\text{Dirac}} \simeq 2.88 \times 10^{-13}.} \quad (427)$$

This small number is not produced by the overlap hierarchy. Therefore the leading PR-II neutrino mass mechanism is identified with the compact Majorana channel:

$$\boxed{\text{neutrino mass} \sim \frac{(v_{\text{EW}}^{\text{PR}})^2}{\bar{M}_{\text{Pl}}} \times \text{determinant lift.}} \quad (428)$$

This does not exclude sterile singlets or mixed scenarios. It states that the primary compact-boundary mass scale generated is Majorana-like.

13.4 Majorana Phase Boundary

Given the PR-II neutrino mass-spectrum candidate, PMNS mixing candidate, and compact Majorana phase rule, the corresponding neutrinoless double-beta effective mass can be evaluated. The numerical result is exact with respect to those stated inputs; the phase relation

$$\phi_M^{\text{PR}} = \pi C_{\text{conf}} \quad (429)$$

remains a compact-boundary closure candidate rather than an independently derived theorem. For the massless anchor ($m_1^{\text{PR}} = 0$), the neutrinoless double-beta effective mass [34, 35] is governed entirely by the interference of the two remaining states:

$$m_{\beta\beta} = \left| m_2 c_{13}^2 s_{12}^2 e^{i\alpha_{21}} + m_3 s_{13}^2 e^{i(\alpha_{31} - 2\delta_{\text{CP}})} \right|. \quad (430)$$

Rather than treating the relative phase shift as a fitted phenomenological unknown, the architecture proposes the compact-boundary relative Majorana interference phase through the confinement response:

$$\phi_M^{\text{PR}} = \pi C_{\text{conf}} = 2.822225357793 \text{ rad} \quad (161.701601836^\circ). \quad (431)$$

Adopting the standard parameterization convention $\alpha_{21} = 0$, the compact-boundary condition fixes the structural interference phase as:

$$\alpha_{31} - 2\delta_{\text{CP}} - \alpha_{21} = \phi_M^{\text{PR}}. \quad (432)$$

Incorporating the previously generated CP-violating phase (δ_{CP}), this phase rule gives the following higher-generation Majorana phase:

$$\alpha_{31}^{\text{PR}} = 3.368400958885 \text{ rad} \quad (192.995158652^\circ). \quad (433)$$

Evaluating the effective mass with these candidate phases gives:

$$m_{\beta\beta}^{\text{PR}} = 1.507694477 \times 10^{-3} \text{ eV}. \quad (434)$$

For this normal-ordering mass-spectrum candidate, arbitrary Majorana [5] phases would permit the envelope $1.410 \text{ meV} \leq m_{\beta\beta} \leq 3.642 \text{ meV}$. The proposed compact-boundary phase selects $m_{\beta\beta}^{\text{PR}} = 1.508 \text{ meV}$, near the lower-interference side of that envelope. The single value is therefore conditional on the phase rule $\phi_M^{\text{PR}} = \pi C_{\text{conf}}$.

13.5 Flavor-Basis Majorana Mass Matrix

Given the neutrino mass-spectrum candidate, PMNS mixing candidate, and Majorana phase rule, the corresponding flavor-basis Majorana mass matrix is

$$M_\nu^{\text{PR}} = U_{\text{PMNS}}^* \text{diag} \left(m_1, m_2 e^{i\alpha_{21}}, m_3 e^{i\alpha_{31}} \right) U_{\text{PMNS}}^\dagger. \quad (435)$$

By construction, this matrix satisfies the Majorana symmetry condition:

$$\left(M_\nu^{\text{PR}} \right)^T = M_\nu^{\text{PR}}. \quad (436)$$

The symmetry $(M_\nu^{\text{PR}})^T = M_\nu^{\text{PR}}$ follows by construction from the Majorana form $U_{\text{PMNS}}^* D U_{\text{PMNS}}^\dagger$. It confirms that the implemented matrix has the required Majorana symmetry, but it does not independently derive the proposed mass eigenvalues or phase rule.

13.6 Input Discipline and Status

The generative boundary for the neutrino mass architecture is restricted. The neutrino spectrum and the associated interference phases are generated exclusively from the pre-established structural ledger; the primary boundary coordinates (q_{bc}, c_{bc}) , the determinant suppression ratio (r_G) , the weak boundary deficit (Δ_{EW}) , the confinement average (C_{conf}) , the previously derived unconfined mixing matrix (U_{PMNS}^{PR}) , and the fundamental dimensional anchor provided by $(v_{EW}^{PR})^2/\bar{M}_{Pl}$.

The architecture derives the masses, mass-squared splittings, and neutrinoless double-beta parameters without any reference to fitted experimental bounds $(m_i, \Delta m_{ij}^2, \sum m_i, m_\beta, m_{\beta\beta})$ or observed phenomenological oscillation angles. These empirical data may enter the framework only post-generation, serving as diagnostic benchmarks to verify the topological projection.

Because the spectrum is generated from the stated boundary rules rather than empirically fitted, the formal structural status of this section is explicitly designated as a:

$$\text{normal-ordering, Majorana-dominant neutrino mass-sector candidate.} \quad (437)$$

Accordingly, the calculated Majorana phase boundary must be interpreted as a structurally derived candidate prediction. It is not an arbitrary phenomenological fit, but the explicit topological consequence of the compact boundary, formally awaiting definitive experimental verification.

13.7 Falsification Bounds

The derivation of the neutrino mass spectrum provides an important boundary for falsification within the framework. Because it operates across a dimension-five topological gap, any structural tampering instantly breaks the predictive closure.

First, the baseline dimension-five topological ratio is required. Stripping the reduced Planck mass suppression factor from the electroweak anchor $((v_{EW}^{PR})^2/\bar{M}_{Pl})$ artificially forces the neutrino spectrum up to the electroweak scale, completely breaking the structural projection.

Second, artificially reversing the determinant hierarchy projection—substituting a geometric suppression (r_G^{+3}) in place of the topological lift (r_G^{-3}) catastrophically fails to generate the heavy atmospheric mass scale (m_3^{PR}) .

Third, omitting the compact neutrino-loss geometric boundary factor $(C_\nu = 1 - \Delta_{EW}c_{bc})$ from the generation chain irreparably distorts the structural mass-squared splittings, pushing the derived ratio entirely outside the predictive closure.

Fourth, the derivation of the terminal mass matrix mathematically relies on the strict unitarity of the unconfined PMNS projection. Furthermore, if the resulting flavor-basis mass matrix fails to satisfy exact symmetry $((M_\nu^{PR})^T = M_\nu^{PR})$, the Majorana-dominant projection is fundamentally inconsistent, immediately invalidating the architecture.

Finally, any derivation that imports experimentally observed target parameters such as neutrino masses (m_i) , phenomenological mass-squared splittings (Δm_{ij}^2) , or observed PMNS mixing angles as generative inputs violates the zero-new-parameter discipline.

Using only the established compact-boundary constants, PR-II must produce a normal-ordered neutrino spectrum, the observed hierarchy of mass splittings, a cosmologically viable total mass, and a definite Majorana interference phase. If matching the data requires fitting any step in this chain, the result no longer qualifies as a zero-new-parameter closure and must instead be treated as a phenomenological model.

To complete the manuscript, we present the assembled final, comprehensive audit map. It explicitly

separates the structural identities from the diagnostic ledgers, formally declaring the no-free-parameter status and the ultimate falsification boundaries of the PR-II manuscript.

14 Audit Map, No-Free-Parameter Status, and Falsification Boundaries

The preceding sections developed the PR-II electroweak and flavor chain from the compact boundary ledger inherited from PR-I. Because the construction produces many numerical outputs, the final task is to separate exact structural identities from closure candidates, precision candidates, and diagnostic ledgers. This section defines that audit map and states the no-free-parameter status of the manuscript.

The central claim is not that every electroweak and flavor result has already been promoted to a final precision theorem. The more precise claim is:

PR-II produces a zero-new-parameter compact-boundary closure candidate for the electroweak and flavor ledger.	(438)
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Target observables are not used as inputs for generation, only as diagnostic references.

14.1 Status Categories

The results in this manuscript are classified into five categories.

1. **Structural identities.** These are algebraic or representation-theoretic results that follow directly from the compact electroweak fiber, such as the $SU(2)_L$ algebra, the residual generator $Q = T_3 + Y/2$, photon masslessness, the tree-level W/Z mass matrix, the Fermi matching identity, neutral-current coupling structure, and anomaly cancellation.
2. **Closure candidates.** These are compact-boundary formulas that generate physically relevant quantities without fitting their measured values. Examples include the weak mixing angle, generation count, strong-coupling anchor, CKM hierarchy, PMNS hierarchy, and neutrino mass-splitting structure.
3. **Precision candidates.** These are closure candidates that numerically agree with reference data at high precision, such as the weak-scale candidate and the charged-lepton precision candidate.
4. **Diagnostic ledgers.** These are required comparison frameworks rather than final predictions. Examples include QCD running, threshold matching, confinement conventions, and radiative electroweak running.
5. **Remaining derivation obligations.** These are compact-boundary rules that currently work as candidates but still require formal uniqueness proofs or radiative/running completion.

This classification is necessary because PR-II spans structural group theory, compact boundary closure, precision numerical candidates, and scheme-dependent QCD diagnostics.

14.2 No-Free-Parameter Input Discipline

To codify the zero-new-parameter constraint of the PR-II architecture, we establish a strict generative audit ledger. The mathematical integrity of the framework relies entirely on separating the foundational inputs (which mathematically drive the topology) from the empirical targets (which serve as post-generation diagnostic references).

Table 17 categorizes the permitted structural inputs and explicitly forbids the importation of target observables [13].

Table 17: PR-II Generative Input Audit Ledger.

Parameter Sector	Specific Variables	Input Status
Fundamental Constants	$\hbar, c, G_N, \bar{M}_{\text{Pl}}$	ALLOWED
Compact-Boundary Data	$q_{\text{bc}}, c_{\text{bc}}, r_G, T_Y, \Delta_{\text{EW}}$	ALLOWED
Group & Topological Data	$C_2^{SU(2)}, C_2^{SU(3)}, Y, T_3, N_{\text{gen}}^{\text{PR}}$	ALLOWED
Electroweak Scale & Bosons	$G_F, v_{\text{EW}}, M_W, M_Z$	FORBIDDEN
Charged Fermion Masses	$m_e, m_\mu, m_\tau, m_u, m_d, m_s, m_c, m_b, m_t$	FORBIDDEN
Flavor-Mixing Matrices	$V_{\text{CKM}}, U_{\text{PMNS}}$	FORBIDDEN
Strong-Force Observables	$\alpha_s(M_Z), \Lambda_{\text{QCD}}$	FORBIDDEN
Neutrino Mass Observables	$\Delta m_{21}^2, \Delta m_{31}^2, \sum_i m_i, m_\beta, m_{\beta\beta}$	FORBIDDEN

The structural derivations presented throughout this manuscript were developed exclusively using the foundational PR variables from the allowed ledger. The physical observables in the forbidden ledger were utilized as external references [14] against which the geometrically generated outputs were evaluated. If any derivation were to import a target observable as a generative input, the zero-new-parameter status would be categorically broken. Such an operation reduces the calculation to a phenomenological fit; it is not a valid PR-II compact-boundary closure.

14.3 Grouped Audit Map

Table 18 summarizes the main outputs of the manuscript and their audit status.

Table 18: Grouped audit map for the PR-II electroweak and flavor chain. Each row lists the controlling equation, generated output, current status, and remaining boundary condition.

Sector	Controlling equation / closure relation	Generated output	Status / remaining boundary
PR-I boundary ledger	$q_{\text{bc}} = 10.905007182855176,$ $c_{\text{bc}} = 0.796684464847899,$ $\alpha_{\text{PR,bc}}^{-1} = 4\pi q_{\text{bc}}$	$\alpha_{\text{PR,bc}}^{-1} = 137.036361812007$	Inherited compact-boundary data from PR-I. Not refit in PR-II.

Continued on next page.

Sector	Controlling equation / closure relation	Generated output	Status / remaining boundary
Electroweak lift	$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L,$ $[T_a, T_b] = i\epsilon_{abc}T_c$	Compact non-Abelian phase fiber and weak projection modes.	Structural identity. Requires genuine $SU(2)_L$, not three Abelian copies.
Projection locking	$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}},$ $Q = T_3 + \frac{Y}{2},$ $Q\Phi_0 = 0$	Residual electromagnetic boundary and standard charge assignments.	Structural identity. Must preserve photon masslessness and charges.
Weak-boson mass ledger	$M_0^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix},$ $\det M_0^2 = 0,$ $M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4},$ $M_Z^2 = \frac{(g^2 + g'^2) v_{\text{EW}}^2}{4}$	$M_\gamma = 0,$ $M_W = M_Z \cos \theta_W,$ $\rho_{\text{EW}} = 1$	Structural tree-level ledger. Precision masses require running and radiative matching.
Weak mixing	$T_Y = \frac{1}{4},$ $\Delta_{\text{EW}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}},$ $\sin^2 \theta_W^{\text{PR}} = T_Y - \Delta_{\text{EW}}$	$\sin^2 \theta_W^{\text{PR}} = 0.231355763298189$	Closure candidate. Remaining task: formal uniqueness of the boundary-deficit rule.
Weak scale and Fermi closure	$\Xi_{\text{EW}} = q_{\text{bc}} c_{\text{bc}} \sin^2 \theta_W^{\text{PR}},$ $\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} e^{-S_{\text{EW}}^{\text{PR}, \text{total}}},$ $v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}},$ $G_F^{\text{PR}} = \frac{1}{\sqrt{2}(v_{\text{EW}}^{\text{PR}})^2}$	$v_{\text{EW}}^{\text{PR}} = 246.219595890 \text{ GeV},$ $G_F^{\text{PR}} = 1.166379220 \times 10^{-5} \text{ GeV}^{-2}$	Precision candidate. Radiative/running precision layer deferred to PR-III.
Compact order mode	$\Phi(x) = \frac{1}{\sqrt{2}} U(x) \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix},$ $4 - 3 = 1$	One scalar order-mode excitation: h_{PR} .	Tree-level scalar-coupling ledger. Scalar mass is deferred to compact Hessian closure in PR-III.
Scalar coupling ledger	$g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}},$ $g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}},$ $g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}$	Tree-level hWW and hZZ couplings, together with an hff coupling conditional on linear order-mode matching.	Structural tree-level ledger. m_h , scalar self-couplings, and longitudinal unitarity amplitudes are PR-III targets.

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Sector	Controlling equation / closure relation	Generated output	Status / remaining boundary
Charged currents	$T^\pm = T_1 \pm iT_2,$ $[Q, T^\pm] = \pm T^\pm,$ $\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} = \frac{1}{2v_{EW}^2}$	Beta-transition topology and low-energy Fermi limit.	Structural identity. Must preserve charged-current normalization and charge bookkeeping.
Neutral currents	$g_L^f = T_3^f - Q_f \sin^2 \theta_W,$ $g_R^f = -Q_f \sin^2 \theta_W,$ $g_V^f = T_3^f - 2Q_f \sin^2 \theta_W,$ $g_A^f = T_3^f$	Tree-level Z -coupling ledger and neutral-current contact structure.	Structural identity. Precision comparison requires electroweak radiative matching.
Fermion representation ledger	$\mathcal{R}_{EW}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2$	Minimal one-generation chiral ledger.	Structural identity. Uses left-handed Weyl convention.
Anomaly cancellation	$[SU(3)_c]^3 = 0,$ $[SU(3)_c]^2 U(1)_Y = 0,$ $[SU(2)_L]^2 U(1)_Y = 0,$ $[U(1)_Y]^3 = 0,$ $[\text{grav}]^2 U(1)_Y = 0,$ $N_2 \in 2\mathbb{Z}$	Gauge and global anomaly consistency.	Structural identity. Failure of any anomaly check breaks the fermion ledger.
Generation origin	$N_{\text{gen}}^{\text{PR}} = \dim \left[\frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3$	Three compact projection sheets.	Closure candidate. Remaining task: formalize projection-sheet interpretation.
Generation hierarchy	$r_G = \frac{c_{bc}}{q_{bc}},$ $\eta_i = r_G^{N_{\text{gen}} - i},$ $y_i^{\text{PR}} = y_T^{\text{PR}} \eta_i$	$y_1 < y_2 < y_3$ with fixed hierarchy ratio q_{bc}/c_{bc} .	Structural flavor ledger. Not yet a mass prediction by itself.
Sector overlap operators	$C_f = C_2^{SU(3)} + C_2^{SU(2)} + \left(\frac{Y_L}{2}\right)^2 + \left(\frac{Y_R}{2}\right)^2,$ $D_f = \kappa_f \text{diag}(y_1, y_2, y_3)$	Charged-lepton, up-quark, down-quark, and neutrino overlap ledgers.	Structural ledger. Requires sector-specific scale/running rules before mass comparison.
Charged leptons	$\theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2} = \frac{2}{9},$ $k_a(\theta_K) = \left[1 + \sqrt{2} \cos \left(\theta_K + \frac{2\pi a}{3} \right) \right]^2,$ $m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a$	$m_e^{\text{PR}} = 0.510984463 \text{ MeV},$ $m_\mu^{\text{PR}} = 105.656418843 \text{ MeV},$ $m_\tau^{\text{PR}} = 1776.934589789 \text{ MeV}$	Precision candidate. Remaining task: formal uniqueness of Koide sheet phase and compact scale corrections.

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Sector	Controlling equation / closure relation	Generated output	Status / remaining boundary
Strong coupling	$(\alpha_3^{\text{PR}})^{-1} = q_{\text{bc}} c_{\text{bc}} - (1 - c_{\text{bc}})$	$\alpha_3^{\text{PR}} = 0.117861507469150$	Closure candidate. Higher-loop and scheme matching remain part of QCD precision layer.
QCD running ledger	$\beta_0(n_f) = 11 - \frac{2}{3}n_f,$ $\alpha_s^{(n_f)}(\mu) = \frac{4\pi}{\beta_0 \ln(\mu^2/\Lambda_{n_f}^2)},$ $\alpha_s^{(n_f)}(\mu_{\text{th}}) = \alpha_s^{(n_f-1)}(\mu_{\text{th}})$	$\Lambda_6, \Lambda_5, \Lambda_4, \Lambda_3$ running and threshold ledger.	Diagnostic ledger. Not a pole-mass theorem; QCD scheme/scale dependence remains.
Heavy-quark thresholds	$\mu_t^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} C_1 C_2,$ $\mu_b^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \frac{r_G}{3} (1 - \Delta_{\text{EW}} c_{\text{bc}}),$ $\mu_c^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} r_G^2 \left(\frac{4}{3}\right) (1 + \Delta_{\text{EW}} A_{\text{EW}})$	$\mu_t^{\text{PR}} = 172.813973267 \text{ GeV},$ $\mu_b^{\text{PR}} = 4.166450483 \text{ GeV},$ $\mu_c^{\text{PR}} = 1.274964814 \text{ GeV}$	Threshold-scale candidates. Not final scheme-independent quark masses.
Light-quark hierarchy	$C_{\text{conf}} = \frac{1 + c_{\text{bc}}}{2},$ $m_s^{\text{PR}} = \mu_b^{\text{PR}} \left(\frac{r_G}{3}\right) (1 + \Delta_{\text{EW}} A_{\text{EW}}) C_{\text{conf}},$ $m_d^{\text{PR}} = m_s^{\text{PR}} r_G c_{\text{bc}} (1 - A_{\text{EW}} \Delta_{\text{EW}}) C_{\text{conf}},$ $m_u^{\text{PR}} = m_d^{\text{PR}} \left(\frac{1 + c_{\text{bc}}}{4}\right) \left(1 + \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4}\right)$	$m_u^{\text{PR}} = 2.146462595 \text{ MeV},$ $m_d^{\text{PR}} = 4.761039493 \text{ MeV},$ $m_s^{\text{PR}} = 94.028510538 \text{ MeV}$	Low-energy running-mass diagnostics. Not pole masses.
CKM mixing	$V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L},$ $\lambda_{\text{PR}} = \sqrt{r_G c_{\text{bc}} C_{\text{conf}} (1 - A_{\text{EW}} \Delta_{\text{EW}})},$ $\delta_{\text{CKM}}^{\text{PR}} = \frac{\pi}{3} + \sqrt{\Delta_{\text{EW}}}$	$ V_{us} = 0.225018430,$ $ V_{cb} = 0.041655455,$ $ V_{ub} = 0.003730932,$ $J_{\text{CKM}}^{\text{PR}} = 3.152605677 \times 10^{-5}$	Closure candidate. Remaining task: prove compact CKM phase-rule uniqueness.
PMNS mixing	$U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L},$ $\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}},$ $\sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}},$ $\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}}(1 + r_G)}{C_{\text{conf}}}$	$\theta_{12}^{\text{PR}} = 33.176356643^\circ,$ $\theta_{23}^{\text{PR}} = 48.735310403^\circ,$ $\theta_{13}^{\text{PR}} = 8.582438059^\circ,$ $\delta_{\text{CP}}^{\text{PR}} = 195.646778408^\circ$	Closure candidate. Must connect to neutrino mass-sector phase structure.
Neutrino masses	$m_1^{\text{PR}} = 0,$ $m_3^{\text{PR}} = \left(\frac{(v_{\text{EW}}^{\text{PR}})^2}{M_{\text{Pl}}}\right) r_G^{-3} c_{\text{bc}} C_\nu,$ $m_2^{\text{PR}} = \sqrt{R_{21}^{\text{PR}}} m_3^{\text{PR}}$	$m_2^{\text{PR}} = 8.627283010 \times 10^{-3} \text{ eV},$ $m_3^{\text{PR}} = 5.010655367 \times 10^{-2} \text{ eV},$ $\Delta m_{21}^{2,\text{PR}} = 7.443001214 \times 10^{-5} \text{ eV}^2,$ $\Delta m_{31}^{2,\text{PR}} = 2.510666721 \times 10^{-3} \text{ eV}^2$	Normal-ordering Majorana-dominant candidate. Majorana phase theorem remains open.
Majorana boundary	$\phi_M^{\text{PR}} = \pi C_{\text{conf}},$ $m_{\beta\beta}^{\text{PR}} = \left m_2 c_{13}^2 s_{12}^2 + m_3 s_{13}^2 e^{i\phi_M^{\text{PR}}} \right $	$m_{\beta\beta}^{\text{PR}} = 1.508 \text{ meV}$	Closure candidate. Experimental test lies beyond current $0\nu\beta\beta$ reach.

This table is intentionally conservative. It prevents structural recovery, precision candidates, and diagnostic comparisons from being conflated.

14.4 Reproducibility Ledger

To enforce arithmetic consistency and verify the strict zero-new-parameter discipline, the PR-II electroweak and flavor construction is fully supported by a test-driven reproducibility ledger. All structural derivations and numerical projections have been independently codified and verified using dedicated Python validation harnesses and Maple algebraic checkers [1]. The complete validation suite, including the evaluation scripts, mathematical outputs, and the generative audit manifest, is publicly available at:

https://github.com/oshetskiresearch/Projection_Relativity

The repository is not a substitute for formal derivation. It documents the stated inputs, reproduces the displayed arithmetic, and checks the implementation for target-input leakage; it does not establish the uniqueness of the proposed closure formulas.

14.5 Remaining Derivation Obligations

The remaining obligations concern formal derivation, uniqueness, and precision matching. They define the work required to test whether the present closure candidates follow uniquely from the compact-boundary architecture or must instead remain independent ansätze. The principal objectives for the next derivation layer are:

1. **Electroweak Boundary Uniqueness:** Determine whether the compact boundary-deficit rule is unique under the stated assumptions and characterize any admissible alternatives.
2. **Non-Abelian Action Integration:** Derive the second-order non-Abelian weak-scale correction explicitly as a deterministic compact-boundary action term to finalize the precision electroweak hierarchy.
3. **Radiative Matching Maps:** Complete the formal electromagnetic and electroweak running layers—specifically the exact $\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z)$ transition—to enable rigorous high-energy precision comparisons.
4. **Generation Topology:** Formalize the precise topological mechanism that defines the projection-sheet origin of the three-generation chiral ledger ($N_{\text{gen}}^{\text{PR}} = 3$).
5. **Charged-Lepton Phase Rigidity:** Derive the proposed Koide sheet phase from an explicit flavor-boundary operator, or retain it as an independent closure ansatz.
6. **QCD Precision Completion:** Finalize the QCD renormalization scheme, running maps, confinement conventions, and threshold-matching prescriptions required to compare the generated compact quark thresholds and light-quark candidates with scheme-specified running and threshold mass definitions.
7. **Flavor Mixing Uniqueness:** Derive CKM and PMNS mixing from explicit PR mass operators and their diagonalizing rotations, formally closing the unitary mixing architecture.
8. **Majorana Phase Theorem:** Derive or falsify the proposed Majorana phase rule from an explicit neutrino-sector operator governing neutrinoless double-beta decay.

These tasks define the PR-III precision and derivation program. Their outcome will determine which PR-II closure rules can be derived from explicit operators or actions, which require modification, and which should remain phenomenological or structural ansätze.

15 Conclusion

Projection Relativity II (PR-II) extends the PR-I compact-boundary architecture into the tree-level electroweak and flavor sectors. The aim is not to replace the Standard Model, but to test whether several quantities normally introduced independently can be organized as outputs of a shared projection framework. The manuscript therefore separates standard structural identities from proposed closure rules, precision candidates, and diagnostic comparisons.

Within this framework, electroweak locking, mass-scale candidates, and flavor constructions are expressed using a common compact-boundary ledger. The manuscript develops the following structural results and closure candidates:

- **The Electroweak Architecture:** Projection locking recovers the tree-level photon and W/Z mass relations, while the weak-mixing and Fermi-scale formulas are presented as compact-boundary closure candidates.
- **The Generation Topology:** A three-generation projection identification is proposed from the three broken electroweak directions, while anomaly cancellation confirms the consistency of the replicated chiral ledger.
- **The Mass and Mixing Ledger:** The stated compact-boundary formulas generate charged-lepton and quark hierarchy candidates together with unitary CKM and PMNS mixing candidates. Derivation from explicit flavor mass operators remains an open task.
- **The Neutrino Sector:** The proposed Planck-suppressed closure rules yield a normal-ordering, Majorana-dominant mass-spectrum candidate and a conditional neutrinoless double-beta effective-mass candidate.

The central methodological constraint is the separation of generation inputs from diagnostic reference values. The displayed candidates are evaluated from the stated compact-boundary rules without inserting the target observables as fitted numerical inputs:

$$\boxed{\text{zero-new-parameter compact-boundary electroweak and flavor closure candidate.}} \quad (439)$$

This manuscript does not claim a radiatively complete precision theory. PR-III is reserved for the derivation and precision tests that remain open: the scalar Hessian, electromagnetic and electroweak running, uniform QCD matching, and the construction of explicit flavor mass operators. Those calculations will determine which PR-II closure candidates survive as consequences of the underlying architecture.

Ultimately, the significance of PR-II lies not in overturning standard gauge theory but in anchoring it. The framework preserves the established currents, anomalies, and effective low-energy structures, while showing that several quantities normally entered independently can be organized as candidate outputs of a shared compact-boundary geometry. If the remaining radiative and running obligations are fulfilled, this architecture would provide a geometric organization for a substantial part of the electroweak and flavor parameter ledger. If those obligations cannot be met, the failure modes identified throughout the manuscript show where the projection program ceases to provide a consistent closure.

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